TRUSTED® COMPUTING GROUP

SPECIFICATION

TCG Storage Opal Family Feature Set: Configurable Locking for NVMe Namespaces and SCSI LUNs

Version 1.30

October 20, 2025

Contact: admin@trustedcomputinggroup.org

PUBLISHED

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CHANGE HISTORY

REVISION	DATE	DESCRIPTION
V1.03/R1.01	February 25, 2022	• Initial revision
V1.30/R1.26	May 28, 2025	• Added informative comments to various sections • Added and clarified terms in the “Definition of Terms” section • Incremented version number from 1.02 to 1.30 • Added an acknowledgement page • Added section titles to interaction cases • Clarified behavior of Pyrite with regards to data encryption • Incorporated work group review comments • Merged with the new feature set template • Added descriptive text to previously empty SP table cells • Incorporated TC review comments

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ACKNOWLEDGEMENT

The TCG wishes to thank all those who contributed to this specification. This document builds on work done in various working groups in the TCG and the industry at large.

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1 Scope

This specification defines the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set.

1.1 Key Words

The key words “MUST,” “MUST NOT,” “REQUIRED,” “SHALL,” “SHALL NOT,” “SHOULD,” “SHOULD NOT,” “RECOMMENDED,” “MAY,” and “OPTIONAL” in this document normative statements are to be interpreted as described in RFC-2119, Key words for use in RFCs to Indicate Requirement Levels.

1.2 Statement Types

Please note a very important distinction between different sections of text throughout this document. There are two distinctive kinds of text: informative comment and normative statements. Because most of the text in this specification will be of the kind normative statements, the authors have informally defined it as the default and, as such, have specifically called out text of the kind informative comment. They have done this by flagging the beginning and end of each informative comment and highlighting its text in gray. This means that unless text is specifically marked as of the kind informative comment, it can be considered a kind of normative statement.

Start of informative comment

This is the first paragraph of 1–n paragraphs containing text of the kind *informative comment* ...

This is the second paragraph of text of the kind *informative comment* ...

This is the nth paragraph of text of the kind *informative comment* ...

To understand the TCG specification the user must read the specification. (This use of MUST does not require any action).

End of informative comment

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2 Introduction and Concepts

2.1 Document Purpose

The Storage Workgroup specifications provide a comprehensive architecture for Storage Devices (SDs) under policy control as determined by the trusted platform host, the capabilities of the SD to conform to the policies of the trusted platform, and the lifecycle state of the SD as a Trusted Peripheral.

2.2 Intended Audience

The intended audience for this specification is both trusted SD manufacturers and developers that want to use these SDs in their systems.

2.3 Conventions

2.3.1 Font Conventions

Names of methods and Security Provider (SP) tables are in `Courier New font` (e.g., the `Set` method, the `Locking` table). This convention does not apply to method and table names appearing in headings or captions.

Hexadecimal numbers are in `Courier New font`.
All other text is in the Arial font.

2.3.2 Security Provider (SP) Table Cell Color Legend

The legend in Table 1 defines the SP table cell color coding, with the RGB values for the shading of each cell indicated in parentheses. This color coding is informative only. The table cell content is normative, unless otherwise indicated. If a table cell is empty or blank (regardless of shading), then the contents of that cell are not specified in this specification. The contents might be specified in another specification.

Start of Informative Comment

As specified in the Core Specification (see [2]) section 3.4.2 and section 5.3.4.3, Access Control limits the methods that can be processed on an SP, a table, or on specific rows and columns of a table. Access Control is specified in layers. The top layer of the mechanism is an Access Control List (ACL). ACLs contain lists of Access Control Elements (ACEs). ACEs contain Boolean combinations of authorities. ACLs and ACEs bind methods to the authorities that are permitted to invoke them. When the state of authorities specified in an ACE Boolean expression are satisfied as described in the Core Specification (see [2]) section 3.4.2.1 (for example, the ACE Boolean expression resolves to TRUE on successful authentication), the ACE permits methods to be invoked on an SP, a table, or on specific rows and columns of a table based on the ACL that contains the ACE.

End of Informative Comment

The “R-W” column in Table 1 describes if the contents of an SP table cell is read-only, writeable, or not required (NR) using color coding. This cell property can be determined through the mapping of methods, that provide read or write access to the cell, and an ACL in the AccessControl table (see Core Specification [2] section 5.3.4.3). If a cell is readable, then there exists a method and associated ACE definition to retrieve the contents of the cell. If a cell is writeable, then there exists a method and associated ACE definition to modify the contents of the cell. There MAY be other conditions that prevent a method from being invoked on a table cell outside the scope of the Access Control specified in the AccessControl table.

The “Value” column in Table 1 describes if the contents of an SP table cell is specified by the feature set, vendor unique (VU), or not required (NR). When a cell content is VU, the cell contents contain the text “VU” and access control for the cell is defined by the feature set.

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The “Access Control” column in Table 1 describes if the access control associated with an SP table cell is user personalizable, fixed, or not defined (ND). Access Control for a cell is personalizable if the Boolean expression of the ACEs associated with the cell are writable. Access Control for a cell is fixed if the Boolean expression of the ACEs associated with the cell are not writeable. Access Control for a cell is not defined (ND) if there does not exist an ACE that is associated with the cell. If the cell contents are (NR), the `Get` method MAY omit the column from a method response. The behavior of `AddACE` and `RemoveACE` methods with respect to ACLs is not defined.

Table 1 – Security Provider (SP) Tables Legend

Table Cell Legend	R-W	Value	Access Control	Comment
Arial-Narrow (230, 230, 230)	Read-only	Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set specified	Fixed	• Cell content is Read-Only. • Access control is fixed. • Value is specified by the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set
Arial Narrow bold-under (230, 230, 230)	Read-only	VU	Fixed	• Cell content is Read-Only. • Access Control is fixed. • Values are Vendor Unique (VU). A minimum or maximum value might be specified.
Arial-Narrow (0, 0, 0)	Not Defined (ND)	Not Required (NR)	Not Defined (ND)	• Cell content is (NR). • Access control is not defined (ND). • Any text in table cell is informative only. • A `Get` MAY omit this column from the method response.
Arial Narrow bold-under (179, 179, 179)	Write	Preconfigured/ user personalizable, or VU	Preconfigured, user personalizable	• Cell content is writable. • Access control is personalizable • `Get` Access Control is not described by this color coding • Cell contents preconfiguration for VU column is “VU”.

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Table Cell Legend	R-W	Value	Access Control	Comment
Arial-Narrow (179, 179, 179)	Write	Preconfigured/ user personalizable, or VU	Fixed	• Cell content is writable. • Access control is fixed. • `Get` Access Control is not described by this color coding • Cell contents preconfiguration for VU column is “VU”.

2.3.3 List Conventions

2.3.3.1 Lists Overview

Lists are associated with an introductory paragraph or phrase, and are numbered relative to that paragraph or phrase (i.e., all lists begin with an a) or 1) entry).

Each item in a list is preceded by an identification with the style of the identification being determined by whether the list is intended to be an ordered list or an unordered list.

If the item in a list is not a complete sentence, the first word in the item is not capitalized. If the item in a list is a complete sentence, the first word in the item is capitalized.

Each item in a list ends with a semicolon, except the last item, which ends in a period. The next to the last entry in the list ends with a semicolon followed by an “and” or an “or” (i.e., “…; and”, or “…; or”). The “and” is used if all the items in the list are required. The “or” is used if only one or more items in the list are required.

2.3.3.2 Unordered Lists

An unordered list is one in which the order of the listed items is unimportant (i.e., it does not matter where in the list an item occurs as all items have equal importance). In an unordered list, each item starts with a lowercase letter followed by a close parenthesis. If it is necessary to subdivide a list item further with an additional unordered list (i.e., have a nested unordered list), then the nested unordered list is indented and each item in the nested unordered list starts with an uppercase letter followed by a close parenthesis.

The following is an example of an unordered list with a nested unordered list:

EXAMPLE - The following are the items for the assembly:
a) a box containing:
    A) a bolt;
    B) a nut; and
    C) a washer;
b) a screwdriver; and
c) a wrench.

2.3.3.3 Ordered Lists

An ordered list is one in which the order of the listed items is important (i.e., item n is required before item n+1). In an ordered list, each item starts with a Western-Arabic numeral followed by a close parenthesis. If it is necessary to subdivide a list item further with an additional unordered list (i.e., have a nested unordered list), then the nested

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unordered list is indented and each item in the nested unordered list starts with an uppercase letter followed by a close parenthesis.

The following is an example of an ordered list with a nested unordered list:

EXAMPLE - The following are the instructions for the assembly:
1) remove the contents from the box;
2) assemble the item;
   A) use a screwdriver to tighten the screws; and
   B) use a wrench to tighten the bolts;
   and
3) take a break.

2.3.4 Numbering Conventions

A binary number is represented in this specification by any sequence of digits consisting of only the Western-Arabic numerals 0 and 1 immediately followed by a lowercase b (e.g., 0101b). Underscores or spaces may be included between characters in binary number representations to increase readability or delineate field boundaries (e.g., 0 0101 1010b or 0_0101_1010b).

A hexadecimal number is represented in this specification by any sequence of digits consisting of only the Western-Arabic numerals 0 through 9 and/or the uppercase English letters A through F immediately preceded by "0x". Underscores or spaces may be included between characters in hexadecimal number representations to increase readability or delineate field boundaries (e.g., 0xFD8C FA23 or 0x0B_FD8C_FA23). Hexadecimal numbers are in Courier New font.

A decimal number is represented in this standard by any sequence of digits consisting of only the Western-Arabic numerals 0 through 9 not immediately followed by a lowercase b or lowercase h (e.g., 25). This standard uses the following conventions for representing decimal numbers:

a) the decimal separator (i.e., separating the integer and fractional portions of the number) is a period;
b) the thousands separator (i.e., separating groups of three digits in a portion of the number) is a space; and
c) the thousands separator is used in both the integer portion and the fraction portion of a number.

A decimal number represented in this standard with an overline over one or more digits following the decimal point is a number where the overlined digits are infinitely repeating (e.g., $666.\overline{6}$ means 666.666 666… or 666 2/3, and $12.\overline{142857}$ means 12.142 857 142 857… or 12 1/7).

2.3.5 Bit Conventions

Name (n:m), where n is greater than m, denotes a set of bits (e.g., Feature (7:0)).

2.3.6 Number Range Conventions

p..q, where p is less than q, represents a range of numbers (e.g., words 100..103 represents words 100, 101, 102, and 103).

2.3.7 Specify and Indicate Conventions

In a given message, command, or other information exchange between a requestor (e.g., a host) and a responder (e.g., an SD):

a) the requestor specifies information in the request; and
b) the responder indicates information in the response.

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2.4 Document References

2.4.1 Document Precedence

If there is a conflict between this specification and any other reference, then the precedence is (where a lower number indicates higher precedence):

1. This specification;
2. Normative references (see section 2.4.2).

Each reference under development and each approved reference may specify its own document precedence.

2.4.2 Normative References

[1] IETF RFC 2119, 1997, "Key words for use in RFCs to Indicate Requirement Levels"

[2] Trusted Computing Group (TCG), "TCG Storage Architecture Core Specification", Version 2.01

[3] Trusted Computing Group (TCG), "TCG Storage Interface Interactions Specification", Version 1.11

[4] Trusted Computing Group (TCG), "TCG Storage Security Subsystem Class: Opal", Version 2.02

[5] Trusted Computing Group (TCG), "TCG Storage Security Subsystem Class: Opalite", Version 1.00

[6] Trusted Computing Group (TCG), "TCG Storage Security Subsystem Class: Ruby", Version 1.00

[7] Trusted Computing Group (TCG), "TCG Storage Security Subsystem Class: Pyrite", Version 2.01

[8] NVM Express Base Specification, Revision 2.1

[9] Trusted Computing Group (TCG), "TCG Storage Opal SSC Feature Set: Single User Mode", Ver 1.00, Rev 2.00

[10] T10 INCITS, "SCSI Architecture Model - 6 (SAM-6)", Revision 08

2.4.3 Informative References

This feature set does not refer to any informative document.

2.5 Dependencies on Other Feature Sets

This feature set does not depend upon any other feature sets.

2.6 Interactions with Other Feature Sets

This feature set defines interactions with the Single User Mode (SUM) Feature Set [9] (see 5.6).

2.7 Definition of Terms

Term	Definition
Configured Locking object	A Locking object where one or more of the following columns has been changed from its default value: RangeStart, RangeLength,

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Term	Definition
	ReadLockEnabled, WriteLockEnabled, ReadLocked, WriteLocked, and LockOnReset
Eradicate	Irrevocably erase (e.g., cryptographically erase)
Global Range Locking Object (GRLO)	The locking object associated with the Global Range. The Global Range Locking object is defined in the Core Specification ([2]) and instantiated in each of the applicable Opal Family SSCs. There is only one Global Range Locking object in the `Locking` table.
Globally-Associated Namespace	Any namespace or LUN that is not associated with a Namespace Global Range Locking object, and is thus associated with the Global Range Locking object
LO	Locking object
LUN	Logical Unit Number
Namespace Global Range Locking object (NGRLO)	For the NVMe interface: A Locking object with a NamespaceID column value not equal to zero or `0xFFFF_FFFF` and a NamespaceGlobalRange column value of TRUE For the SCSI interface: A Locking object with a NamespaceID column value not equal to `0xFFFF_FFFF` and a NamespaceGlobalRange column value of TRUE Namespace Global Range Locking object is defined in the overview section of this specification (section 3.1)
Namespace Non-Global Range Locking object (NNGRLO)	For the NVMe interface: A Locking object with a NamespaceID column value not equal to zero or `0xFFFF_FFFF` and a NamespaceGlobalRange column value of FALSE For the SCSI interface: A Locking object with a NamespaceID column value not equal to `0xFFFF_FFFF` and a NamespaceGlobalRange column value of FALSE Namespace Non-Global Range Locking object is defined in the overview section of this specification (section 3.1)
Namespace/LUN	NVMe Namespace or SCSI LUN (Logical Unit Number)
Non-Global Range Locking object (Non-GRLO)	Any Locking object other than the Global Range Locking object
ND	Not Defined
NR	Not Required
NS	Namespace

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Term	Definition
OFS	Original Factory State
Opal Family	Any of: Opal SSC [4], Opalite SSC [5], Ruby SSC [6], or Pyrite SSC [7]
Read Locked	The state of a Locking object with a ReadLockEnabled column value of TRUE and a ReadLocked column value of TRUE
Read Unlocked	The state of a Locking object with a ReadLockEnabled column value of FALSE or a ReadLocked column value of FALSE
SD	Storage Device
SP	Security Provider
SSC	Security Subsystem Class. See elaborate definition in [2]
SUM	Single User Mode. See [9]
TPer	Trusted Peripheral. The Trusted Peripheral is the subset of a Storage Device for which TCG manages security
Unused Key Count	The number of media encryption key resources which are not in use and are available for use (see section 5.2.1.9)
Write Locked	The state of a Locking object with a WriteLockEnabled column value of TRUE and a WriteLocked column value of TRUE
Write Unlocked	The state of a Locking object with a WriteLockEnabled column value of FALSE or a WriteLocked column value of FALSE

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3 Namespaces/LUNs Overview

3.1 Namespace/LUN Operations Overview

The Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set provides a means for a Locking SP to support separate locking management of different namespaces/LUNs, as well as of LBA ranges within a namespace/LUN.

Start of Informative Comment

This specification uses the term Namespace/LUN to reflect that its functionality applies to both NVMe namespaces and SCSI LUNs. For conciseness and backward compatibility, the names of method parameters and table columns are restricted to the term Namespace and its derivatives; these parameters and columns refer to both NVMe namespaces and SCSI LUNs.

End of Informative Comment

This specification defines the following Locking object usages:

a) A Namespace Global Range Locking object is the first Locking object to be uniquely associated with a namespace/LUN; and

b) A Namespace Non-Global Range Locking object is a Locking object associated with an LBA range within a namespace/LUN.

This specification also defines the relationships of those Locking objects to LBA ranges, the interactions with the Global Range Locking object, and interactions with interface commands.

This specification defines two new methods, `Assign` (see 4.1.1.1) and `Deassign` (see 4.1.1.2). These methods associate Locking objects with namespaces/LUNs and, optionally, with LBA ranges in namespaces/LUNs.

This specification adds two columns to the `Locking` table, NamespaceID (see 4.2.2.1.1) and NamespaceGlobalRange (see 4.2.2.1.2). The column values determine with which namespace/LUN a Locking object is associated and whether a Locking object is a Namespace Global Range Locking object or a Namespace Non-Global Range Locking object.

Each logical block is associated with exactly one Locking object.

A Globally-Associated Namespace is a namespace/LUN that is associated with the Global Range Locking object.

If the Storage Device supports the NVMe interface, then the Locking SP is in one of three modes:

a) Global LO / Multiple NS: The TPer contains one or more namespaces, all of which are associated many-to-one with the Global Range Locking object;

b) Multiple LO / Single NS: The TPer contains exactly one namespace, which is associated one-to-one with the Global Range Locking object. LBA ranges in the namespace MAY be associated one-to-one with Non-Global Range Locking objects, by invoking the `Set` method; or

c) Multiple LO / Multiple NS: The TPer contains one or more namespaces, each of which is associated either many-to-one with the Global Range Locking object, or using the `Assign` method one-to-one with a Namespace Global Range Locking object. Each LBA range of a namespace MAY be associated one-to-one with a Namespace Non-Global Range Locking object.

If the Storage Device supports the SCSI interface, then the Locking SP is in one of two modes:

a) Global LO / Multiple NS: The device contains one or more LUNs, all of which are associated many-to-one with the Global Range Locking object; or

b) Multiple LO / Multiple NS: The device contains one or more LUNs, each of which is associated either many-to-one with the Global Range Locking object, or using the `Assign` method one-to-one with a Namespace

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Global Range Locking object. Each LBA range of a LUN MAY be associated one-to-one with a Namespace Non-Global Range Locking object.

These modes are shown in Figure 1 and Figure 2.

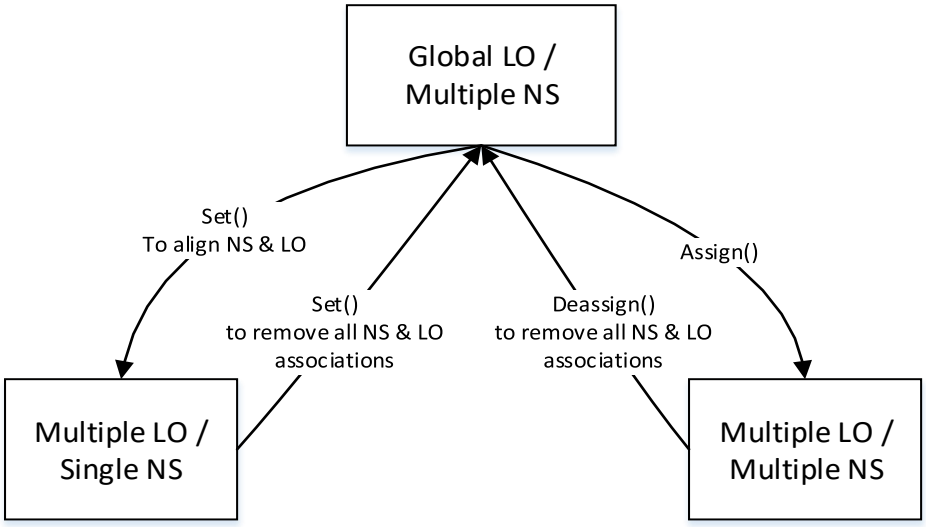


Figure 1 - Locking SP Modes (NVMe)

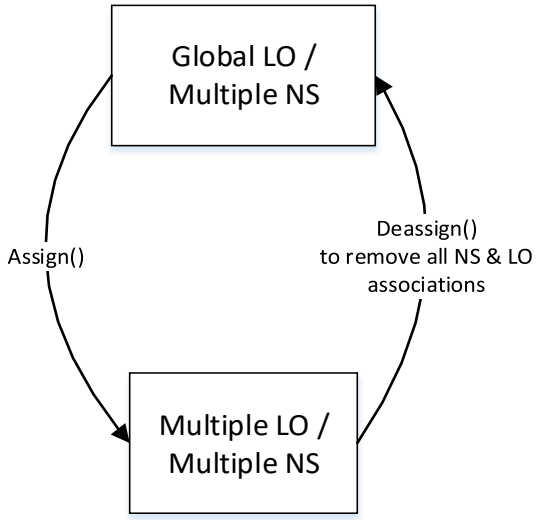


Figure 2 - Locking SP Modes (SCSI)

The uses of the `Set` method defined in [3] perform the transitions between “Global LO / Multiple NS mode” and “Multiple LO / Single NS mode” as shown above (although the terms “Global LO / Multiple NS mode” and “Multiple LO / Single NS mode” are not used in [3]). Transitions between the Global LO / Multiple NS mode and the Multiple LO / Multiple NS mode are specified in this specification. Transitions directly between the Multiple LO / Single NS mode and the Multiple LO / Multiple NS mode are not allowed.

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If the Storage Device supports the NVMe interface and exactly one namespace exists in the TPer, and if no Non-Global Range Locking objects have been configured (see definition of Configured Locking Object), then the TPer is in the Global LO / Multiple NS mode.

If the Storage Device supports the NVMe interface and exactly one namespace exists in the TPer, and if at least one Non-Global Range Locking object has been configured using the `Set` method, then the TPer is in the Multiple LO / Single NS mode.

In Multiple LO / Single NS mode, requirements are specified in [3] and the only section of this specification that applies is Level 0 Discovery (see 5.2).

In Multiple LO / Multiple NS mode, the requirements in [3] for single namespaces and multiple namespaces are overridden by the requirements in this specification.

3.2 Interactions with the Global Range Locking Object

3.2.1 Errors reported for commands that would violate the locked state of the GRLO

Start of Informative Comment

The following rules specify the errors reported for commands that would violate the Read Locked or Write Locked state of a Globally-Associated namespace.

End of Informative Comment

If the Global Range Locking object is Read Locked, then any command that attempts to read user data or metadata in a Globally-Associated Namespace SHALL fail with a status of Data Protection Error. See [3] for more information.

If the Global Range Locking object is Write Locked, then any command that attempts to modify user data or metadata in a Globally-Associated Namespace SHALL fail with a status of Data Protection Error. See [3] for more information.

3.2.2 Operations on the media encryption keys of the GRLO

Start of Informative Comment

The following rules specify operations on the media encryption keys of Globally-Associated Namespaces.

End of Informative Comment

The TPer SHALL support a minimum of one media encryption key per namespace. In this case, the K_AES_* object referenced by the ActiveKey column value of the Global Range Locking object SHALL be a collective representation of all the media encryption keys in use for Globally-Associated Namespaces. Any method that modifies the Key column of the K_AES_* object indicated by the ActiveKey column of the Global Range Locking object SHALL be applied individually to each of the keys represented by that K_AES_* object. Successful execution of any method that results in the cryptographic erase of the Global Range Locking object SHALL result in the cryptographic erase of all Globally-Associated Namespaces.

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3.3 NVMe Interactions

Start of Informative Comment

The following section describes interactions between the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set and the NVMe interface.

End of Informative Comment

3.3.1 Examples of Interactions with SIIS Single Namespace and Multiple Namespace Specifications

Start of Informative Comment

Table 2 illustrates a Locking SP in which all namespaces are implicitly associated with the Global Range Locking object.

Table 2 - Example: Global LO / Multiple NS

Locking Object	Namespace
Global	NS1 NS2 ... NSn

Table 3 illustrates a Locking SP with only a single namespace, but where invocations of the `Set` method have associated eight LBA ranges of the namespace with Non-Global Range Locking objects. The number of each LBA range indicates the order in which that range was associated with a Locking object, e.g., NS1 LBA Range 1 was associated with LO1 first, NS1 LBA Range 2 was associated with LO2 second, etc. See [3].

Table 3 - Example: Multiple LO / Single NS

Locking Object	Namespace	Range within Namespace
Global	NS1	NS1 Global
LO1	NS1	NS1 LBA Range 1
LO2	NS1	NS1 LBA Range 2
LO3	NS1	NS1 LBA Range 8
LO4	NS1	NS1 LBA Range 4
LO5	NS1	NS1 LBA Range 6
LO6	NS1	NS1 LBA Range 5
LO7	NS1	NS1 LBA Range 7
LO8	NS1	NS1 LBA Range 3

Table 4 illustrates a Locking SP in which invocations of the `Assign` method have associated each of four namespaces with a different Non-Global Range Locking object. Three other namespaces remain implicitly associated with the Global Range Locking object.

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Table 4 - Example: Multiple LO / Multiple NS

Locking Object	Namespace	Range within Namespace
Global	NS1 NS3 NS5	NS1 Global NS3 Global NS5 Global
LO1	NS6	NS6 Global
LO2	NS6	NS6 LBA Range 1
LO3	NS2	NS2 Global
LO4	NS4	NS4 Global
LO5	NS7	NS7 Global
LO6	NS7	NS7 LBA Range 2
LO7	NS7	NS7 LBA Range 1

The association of a namespace and its LBA ranges with one or more Locking objects is shown in Figure 3. When a namespace is first created, it is associated by default with the Global Range Locking object. When the `Assign` method is first invoked on a namespace, it selects a Non-Global Range Locking object in the `Locking` table with a NamespaceID column value of zero (i.e., is not associated with a namespace), configures the new Locking object as a Namespace Global Range Locking object, and associates the namespace with the new Locking object.

When the `Assign` method is invoked on a namespace which is associated with a Namespace Global Range Locking object, it selects a Non-Global Range Locking object in the `Locking` table with a NamespaceID column value of zero, configures the new Locking object as a Namespace Non-Global Range Locking object, and associates the specified LBA range of that namespace with the new Locking object. Multiple LBA ranges within a namespace may be associated with different Locking objects. LBA ranges that have not been associated with Namespace Non-Global Range Locking objects are by default associated with the Namespace Global Range Locking object.

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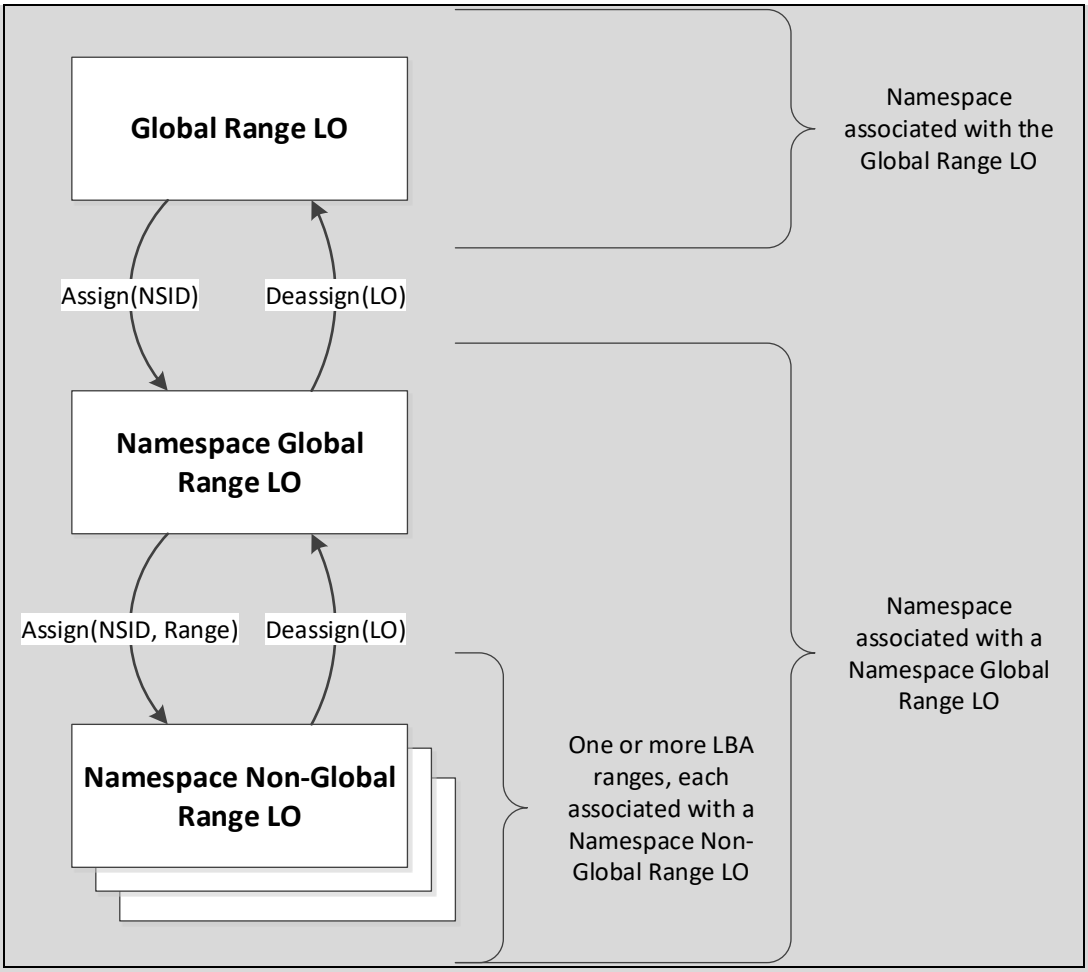


Figure 3 - Flows in Namespace Associations with Locking Objects

Invocation of the `Deassign` method on a Namespace Non-Global Range Locking object returns the LBA range associated with the Namespace Global Range Locking object. When there are no Namespace Non-Global Range Locking objects associated with a namespace, then the `Deassign` method may be invoked on the Namespace Global Range Locking object. This causes the namespace (and implicitly all its LBA ranges) to be associated with the Global Range Locking object.

End of Informative Comment

3.3.2 Interactions with the Namespace Management Command

The Namespace Management command (see [8]) MAY be supported on a TPer that supports the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set.

3.3.2.1 Normal operation of namespace creation

Start of Informative Comment

The following rule specifies normal operation of namespace creation.

End of Informative Comment

If:

a) the Select (SEL) field of the command is Create;

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b) the Global Range Locking object is Read Unlocked;
c) the Global Range Locking object is Write Unlocked; and
d) the Unused Key Count is greater than or equal to one,

then:

1) the Namespace Management command SHALL be processed as defined in [8]; and
2) if the Namespace Management command succeeds in any life cycle state of the SPs, then:
    a. the Unused Key Count SHALL be decremented by one; and
    b. the new namespace SHALL be associated with an unused media encryption key.

3.3.2.2 Preventing namespace creation if no media encryption key is available

Start of Informative Comment

The following rule prevents creation of a namespace if there is no media encryption key resource available.

End of Informative Comment

If:

a) the Select (SEL) field of the command is Create;
b) the Global Range Locking object is Read Unlocked;
c) the Global Range Locking object is Write Unlocked;
d) the command would otherwise succeed; and
e) the Unused Key Count is zero,

then:

a) the Namespace Management command SHALL fail with a status of Operation Denied; and
b) the Unused Key Count SHALL NOT be changed.

3.3.2.3 Preventing namespace creation if the Global Range Locking Object is locked

Start of Informative Comment

The following rule prevents creation of a namespace if the Global Range Locking Object is Read Locked or Write Locked.

End of Informative Comment

If:

a) the Select (SEL) field of the command is Create;
b) the Unused Key Count is greater than or equal to one;
c) the command would otherwise succeed; and
d) the Global Range Locking object is:
    a. Read Locked; or
    b. Write Locked,

then:

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a) the Namespace Management command SHALL fail with a status of Operation Denied; and

b) the Unused Key Count SHALL NOT be changed.

3.3.2.4 Normal operation of namespace deletion

Start of Informative Comment

The following rule specifies normal operation of namespace deletion. For the Namespace Management command to succeed, all affected namespaces must be associated with the Global Range Locking object.

End of Informative Comment

If:

a) the Select (SEL) field of the command is Delete;

b) the Global Range Locking object is Read Unlocked;

c) the Global Range Locking object is Write Unlocked; and

d) the Namespace Identifier (NSID) field of the command does not specify any namespace associated with a Namespace Global Range Locking object;

then:

1. the Namespace Management command SHALL be processed as defined in [8]; and
2. if the Namespace Management command succeeds in any life cycle state of the SPs, then:
    a. the Unused Key Count SHALL be incremented by the number of namespaces that were deleted; and
    b. the media encryption key associated with any deleted namespace SHALL be eradicated.

3.3.2.5 Preventing namespace deletion if the namespace is associated with an NGRLO

Start of Informative Comment

The following rule prevents deletion of a namespace that is associated with a Namespace Global Range Locking Object.

End of Informative Comment

If:

a) the Select (SEL) field of the command is Delete;

b) the Global Range Locking object is Read Unlocked;

c) the Global Range Locking object is Write Unlocked:

d) the command would otherwise succeed; and

e) the Namespace Identifier (NSID) field of the command specifies one or more namespaces associated with a Namespace Global Range Locking object;

then the Namespace Management command SHALL fail with a status of Operation Denied.

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3.3.2.6 Preventing namespace deletion if the namespace is associated with the GRLO and the GRLO is locked

Start of Informative Comment

The following rule prevents deletion of a namespace associated with the Global Range Locking object if the Global Range Locking object is Read Locked or Write Locked.

End of Informative Comment

If:

a) the Select (SEL) field of the command is Delete;

b) the Namespace Identifier (NSID) field of the command does not specify any namespace associated with a Namespace Global Range Locking object;

c) the command would otherwise succeed; and

d) the Global Range Locking object is:

   a. Read Locked; or

   b. Write Locked,

then the Namespace Management command SHALL fail with a status of Operation Denied.

3.3.3 Interactions with the Format NVM Command

Start of Informative Comment

The Format NVM command specifies that either one namespace or all namespaces (i.e., namespace ID `0xFFFF_FFFF`) are to be formatted.

End of Informative Comment

3.3.3.1 Behavior of Format NVM Command if none of the affected LBAs are write-locked

If the Format NVM command is received by the Storage Device and all logical blocks in every targeted Namespace are associated with a Locking object that is in a Write Unlocked state, then the Format NVM command SHALL be processed as specified in [8] and SHALL NOT alter the Locking SP configuration, including PIN values, access control settings, etc. See section 3.1 for more details on LBA and logical block association to Locking objects.

3.3.3.2 Behavior of Format NVM Command if any of the LBAs are write-locked

If the Format NVM command is received by the Storage Device and any logical block in any targeted Namespace is associated with a Locking object that is in a Write Locked state, then the Format NVM command SHALL fail with a status of Invalid Security State. See section 3.1 for more details on LBA and logical block association to Locking objects.

3.4 SCSI Interactions

Start of Informative Comment

The following section describes interactions between the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set and the SCSI interface.

End of Informative Comment

3.4.1 Interactions with the SCSI LUNs

Start of Informative Comment

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The following rules specify configuration requirements for SCSI LUNs (see [10] for details).

End of Informative Comment

The Storage Device SHALL support a minimum of 2 LUNs.

The Storage Device SHALL NOT support LUN ID values that cannot be independently addressed using only the first four bytes of the LUN ID as described in section 4.2.2.1.1.

The Storage Device SHALL NOT support a LUN ID that maps to a NamespaceID column value of `0xFFFF_FFFF` as described in section 4.2.2.1.1.

3.4.2 Interactions with the SCSI Commands that affect multiple LUNs

Start of Informative Comment

SCSI commands may affect one or more LUNs on the Storage Device (see [10] for details).

The following rules specify the behavior when a command affects multiple LUNs.

End of Informative Comment

3.4.2.1 Behavior of Read Commands if any of the associated locking objects are read-locked

If:

a) the command affects two or more LUNs;

b) the command is a Read Command (see [3]); and

c) one or more of the Locking objects associated with any of the affected LUNs is Read Locked,

then the command SHALL behave as defined in the SCSI command interactions with the Locking SP table (see [3]).

3.4.2.2 Behavior of Write Commands if any of the associated locking objects are write-locked

If:

a) the command affects two or more LUNs;

b) the command is a Write Command (see [3]); and

c) one or more of the Locking objects associated with any of the affected LUNs is Write Locked,

then the command SHALL behave as defined in the SCSI command interactions with the Locking SP table (see [3]).

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4 Security Subsystem Class (SSC) Specific Functionality

This section specifies the additional SSC-specific functionality (not contained in the Core Specification [2] or in the applicable Opal Family SSC) required to support the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set.

Note that Pyrite storage devices' behavior regarding parts of this specification related to media encryption and encryption keys management is undefined. The reason is that the Pyrite SSC specification [7] does not require encryption support.

4.1 Methods

This section defines new methods and modifications to existing methods required for this feature set.

4.1.1 New Methods

This section defines the new methods that are required to support this feature set.

4.1.1.1 Assign (M)

The `Assign` method is a Locking Template-specific method.

Start of Informative Comment

The `Assign` method selects a Non-Global Range Locking object in the `Locking` table that has a NamespaceID column value of zero or `0xFFFF_FFFF` (i.e., that is not associated with a namespace/LUN), and sets multiple column values in a single operation. The `Assign` method returns the UID column value of the selected Locking object and the NamespaceGlobalRange column value.

Following a successful invocation of the `Assign` method, the Namespace, or the specified LBA range within the Namespace, is governed by the properties of the selected Non-Global Range Locking object.

The `Assign` method is invoked upon a Locking SP that is in either the Global LO / Multiple NS mode or the Multiple LO / Multiple NS mode (see Figure 1).

End of Informative Comment

The following pseudo-code is the signature of the `Assign` method (see [2] for more information).

```
LockingTableUID.Assign [
     NamespaceID : bytes_4,
     RangeStart     = uinteger_8,
     RangeLength  = uinteger_8,
     AssignToSUMRange = boolean]
=>
[ UID                  : uidref,
  NamespaceGlobalRange : boolean ]
```

Method UID: 00 00 00 06 00 00 08 04

4.1.1.1.1 Parameter Descriptions

4.1.1.1.1.1 NamespaceID

The NamespaceID parameter specifies the value to which the NamespaceID column of the Locking object (see 4.2.2.1.1) SHALL be set. The bytes-ordering of the NamespaceID parameter is Big-Endian.

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4.1.1.1.1.2 RangeStart

The RangeStart parameter (if present) specifies the value to which the RangeStart column of the Locking object SHALL be set.

4.1.1.1.1.3 RangeLength

The RangeLength parameter (if present) specifies the value to which the RangeLength column of the Locking object SHALL be set.

4.1.1.1.1.4 AssignToSUMRange

Start of Informative Comment

Per [9], some locking ranges may be activated in SUM and others may not. The AssignToSUMRange parameter allows the host to tell the SD that the namespace should be assigned to a locking range activated in Single User Mode or not.

A note to system integrators: If a host software does not implement this parameter (due to supporting version 1.01 or earlier of this specification), then the host software cannot assign a SUM locking object.

End of Informative Comment

The AssignToSUMRange parameter, if present and set to TRUE, specifies that the SD SHALL assign the namespace to a Locking object that is included in the SingleUserModeRanges column of the `LockingInfo` table. See section 5.6.2.1.

4.1.1.1.2 Returned Value Descriptions

4.1.1.1.2.1 UID

The UID value is the UID column value of the selected Locking object.

4.1.1.1.2.2 NamespaceGlobalRange

The NamespaceGlobalRange value is the NamespaceGlobalRange column value of the selected Locking object (see 4.2.2.1.2).

4.1.1.1.3 Assign Method Operation

The operation of the `Assign` method depends in part upon whether a Locking object is associated with the namespace/LUN specified by the method and whether the Storage Device supports Namespace Non-Global Range Locking objects:

a) If the `Locking` table contains no Locking objects associated with the specified namespace, then `Assign` configures the Namespace Global Range Locking object for that namespace; and

b) If the `Locking` table contains a Namespace Global Range Locking object associated with the specified namespace and the RANGE_C bit is set to one in the Configurable Namespace Locking Feature Descriptor (see 5.2.1), then `Assign` configures a Namespace Non-Global Range Locking object for that namespace. If the RangeLength parameter is set to zero, then no logical blocks are associated with that Locking object. Subsequently, the `Set` method MAY be invoked to associate logical blocks with that Locking object.

In order to prevent invalid configurations of the Locking SP, the following uses of `Assign` are not allowed:

a) configuring a Namespace Non-Global Range Locking object in a Storage Device that does not support Non-Global Range Locking objects; and

b) configuring a Namespace Non-Global Range Locking object for a range of a namespace that overlaps a range specified by another Namespace Non-Global Range Locking object for the same namespace.

The invocation of the `Assign` method (see 4.1.1.1) on a Globally-Associated Namespace associates a Namespace Global Range Locking object with that namespace/LUN. If `Assign` is invoked again on that namespace, then it

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associates a Namespace Non-Global Range Locking object with a (possibly zero-length) range of LBAs within that namespace/LUN.

If a namespace/LUN has LBAs that are not associated with any of the Namespace Non-Global Range Locking objects for that namespace/LUN, then those LBAs are associated with the Namespace Global Range Locking object.

A Namespace Non-Global Range Locking object with a RangeLength column value of zero is considered to not overlap with any other LBA range, regardless of the RangeStart column value.

4.1.1.1.3.1 Assigning a Namespace Global Range Locking object (NGRLO)

4.1.1.1.3.1.1 Assignment of an NGRLO via the NVMe Interface

Start of Informative Comment

The following rule specifies successful operation of the `Assign` method when assigning a Namespace Global Range Locking object on Storage Devices that support the NVMe interface.

User data residing in the specified namespace's LBA range continues to be accessible after the `Assign` method completes.

Note: The behavior of the `Assign` method as described in items 1 and 2 below depends on the SD's support for Single User Mode Feature Set (see [9]). A SD would not need to implement both items 1 and 2, but instead one or the other.

End of Informative Comment

If:

a) the Storage Device supports the NVMe interface

b) the NamespaceID parameter:

   a. is not equal to zero;

   b. is not equal to `0xFFFF_FFFF`; and

   c. specifies a value that is not equal to the NamespaceID column value in any Locking object;

c) the RangeStart parameter, if present, is set to zero;

d) the RangeLength parameter, if present, is set to zero;

e) the Global Range Locking object is Write Unlocked;

f) the Global Range Locking object is Read Unlocked; and

g) the AssignToSUMRange parameter is not specified or is set to FALSE,

then the `Assign` method SHALL:

1. If the SD supports the Single User Mode Feature Set (see [9]), select a Non-Global Range Locking object in the `Locking` table with:

   a. a NamespaceID column value of zero; and

   b. a UID column value that is not included in the SingleUserModeRanges column of the `LockingInfo` table;

2. If the SD does not support the Single User Mode Feature Set (see [9]), select a Non-Global Range Locking object in the `Locking` table with a NamespaceID column value of zero.

3. set the NamespaceID column value to the value of the NamespaceID parameter;

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4. set the RangeStart column value to zero;
5. set the RangeLength column value to zero;
6. set the NamespaceGlobalRange column value to TRUE;
7. keep the current Unused Key Count;
8. transfer control of the media encryption key associated with the namespace to the K_AES_* object indicated by the ActiveKey column value of the selected Locking object; and
9. return:
    a) the UID of the selected Locking object;
    b) the NamespaceGlobalRange column value; and
    c) a status code of SUCCESS.

Assigning a Namespace Global Range Locking object SHALL NOT cause cryptographic erase of that namespace.

4.1.1.1.3.1.2 Assignment of an NGRLO via the SCSI Interface

Start of Informative Comment

The following rule specifies successful operation of the `Assign` method when assigning a Namespace Global Range Locking object on Storage Devices that support the SCSI interface.

User data residing in the specified LUN continues to be accessible after the `Assign` method completes.

Note: The behavior of the `Assign` method as described in items 1 and 2 below depends on the SD's support for Single User Mode Feature Set. A SD would not need to implement both items 1 and 2, but instead one or the other.

End of Informative Comment

If:

a) the Storage Device supports the SCSI interface;
b) the NamespaceID parameter:
    a. is not equal to `0xFFFF_FFFF`; and
    b. specifies a value that is not equal to the NamespaceID column value in any Locking object;
c) the RangeStart parameter, if present, is set to zero;
d) the RangeLength parameter, if present, is set to zero;
e) the Global Range Locking object is Write Unlocked;
f) the Global Range Locking object is Read Unlocked; and
g) the AssignToSUMRange parameter is not specified or is set to FALSE,

then the `Assign` method SHALL:

1. If the SD supports the Single User Mode Feature Set (see [9]), select a Non-Global Range Locking object in the `Locking` table with:
    a. a NamespaceID column value of zero; and

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b. a UID column value that is not included in the SingleUserModeRanges column of the `LockingInfo` table;

2. If the SD does not support the Single User Mode Feature Set (see [9]), select a Non-Global Range Locking object in the `Locking` table with a NamespaceID column value of zero. set the NamespaceID column value to the value of the NamespaceID parameter;
3. set the RangeStart column value to zero;
4. set the RangeLength column value to zero;
5. set the NamespaceGlobalRange column value to TRUE;
6. keep the current Unused Key Count;
7. transfer control of the media encryption key associated with the LUN to the K_AES_* object indicated by the ActiveKey column value of the selected Locking object; and
8. return:
    a. the UID of the selected Locking object;
    b. the NamespaceGlobalRange column value; and
    c. a status code of SUCCESS.

Assigning a Namespace Global Range Locking object SHALL NOT cause cryptographic erase of that LUN.

4.1.1.1.3.1.3 Preventing the assignment of an NGRLO if the GRLO is locked

Start of Informative Comment

The following rule prevents assignment of a Namespace Global Range Locking object if the Global Range Locking object is Write Locked or Read Locked.

End of Informative Comment

If:

a) the Storage Device supports the NVMe interface

b) the NamespaceID parameter:
    a. is not equal to zero;
    b. is not equal to `0xFFFF_FFFF`; and
    c. specifies a value that is not equal to the NamespaceID column value in any Locking object;

    and

c) the Global Range Locking object is Write Locked or Read Locked,

then the `Assign` method SHALL fail with a status of FAIL.

If:

a) the Storage Device supports the SCSI interface

b) the NamespaceID parameter:

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a. is not equal to `0xFFFF_FFFF`; and

b. specifies a value that is not equal to the NamespaceID column value in any Locking object; and

c) the Global Range Locking object is Write Locked or Read Locked,

then the `Assign` method SHALL fail with a status of FAIL.

4.1.1.1.3.1.4 Preventing the assignment of an NGRLO if it specifies an LBA range

Start of Informative Comment

The following rule prevents assignment of a Namespace Global Range Locking object that specifies an LBA range.

End of Informative Comment

If:

a) the Storage Device supports the NVMe interface

b) the NamespaceID parameter:

a. is not equal to zero;

b. is not equal to `0xFFFF_FFFF`; and

c. specifies a value which is not equal to the NamespaceID column value in any Locking object; and

c) other parameters are set as follows:

a. the RangeStart parameter is set to a nonzero value; or

b. the RangeLength parameter is set to a nonzero value,

then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

If:

a) the Storage Device supports the SCSI interface

b) the NamespaceID parameter:

a. is not equal to `0xFFFF_FFFF`; and

b. specifies a value that is not equal to the NamespaceID column value in any Locking object; and

c) other parameters are set as follows:

a. the RangeStart parameter is set to a nonzero value; or

b. the RangeLength parameter is set to a nonzero value,

then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

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4.1.1.1.3.2 Assigning a Namespace Non-Global Range Locking object (NNGRLO)

Start of Informative Comment

The following rule specifies successful operation of the `Assign` method when assigning a Namespace Non-Global Range Locking object on Storage Devices that support the NVMe interface.

The `Assign` method, when specifying a Namespace Non-Global Range Locking object, transitions an LBA range to the control of that locking object. Since the media encryption key used for the LBA range has changed (to the one used by that locking object), all data written to this LBA range prior to the transition becomes inaccessible.

For an LBA range associated with a non-SUM Namespace Global Range Locking object that is written, then assigned to a non-SUM Namespace Non-Global Range Locking object, and then deassigned and returned to its former locking object's control without any additional writes, this feature-set does not specify whether such written data is unrecoverable.

Note: The behavior of the `Assign` method as described in items 1 and 2 below depends on the SD's support for Single User Mode Feature Set (see [9]). A SD would not need to implement both items 1 and 2, but instead one or the other

End of Informative Comment

4.1.1.1.3.2.1 Assignment of an NNGRLO via the NVMe Interface

If:

a) the Storage Device reports a value of one in the Range Capable (Range_C) field of the Configurable Namespace Locking Feature Descriptor (see 5.2.1);

b) the NamespaceID parameter specifies a value that is equal to the NamespaceID column value in a Namespace Global Range Locking object;

c) the RangeStart parameter and the RangeLength parameter, if present, specify an LBA range in which no logical blocks are assigned to any Namespace Non-Global Range Locking object having a NamespaceID column value equal to the NamespaceID parameter;

d) the Maximum Ranges Per Namespace field of the Configurable Namespace Locking Feature Descriptor (see 5.2.1) specifies:

   a. a value of `0xFFFF_FFFF`; or

   b. a value less than `0xFFFF_FFFF` and that value is greater than the number of Namespace Non-Global Range Locking objects that have a NamespaceID column value equal to the NamespaceID parameter;

e) the Unused Key Count is greater than or equal to one; and

f) the AssignToSUMRange parameter is not specified or is set to FALSE,

then the `Assign` method SHALL:

1. If the SD supports the Single User Mode Feature Set (see [9]), select a Non-Global Range Locking object in the `Locking` table with:

   a. a NamespaceID column value of zero or `0xFFFF_FFFF`;

   b. a UID column value that is not included in the SingleUserModeRanges column of the `LockingInfo` table;

2. If the SD does not support the Single User Mode Feature Set (see [9]), select a Non-Global Range Locking object in the `Locking` table with a NamespaceID column value of zero or `0xFFFF_FFFF`;

3. set the NamespaceID column value to the value of the NamespaceID parameter;

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4. set the RangeStart column value to the specified value or to zero if not specified;
5. set the RangeLength column value to the specified value or to zero if not specified;
6. set the NamespaceGlobalRange column value to FALSE;
7. decrement the Unused Key Count by one; and
8. return:
    a. the UID of the selected Locking object;
    b. the NamespaceGlobalRange column value; and
    c. a status code of SUCCESS.

4.1.1.1.3.2.2 Preventing the assignment of an NNGRLO if the storage device does not support NNGRLO

Start of Informative Comment

The following rule prevents assignment of a Namespace Non-Global Range Locking object if the Range Capable bit is set to 0 in the Level 0 Discovery response data.

End of Informative Comment

If:

a) the Storage Device reports a value of zero in the Range Capable (Range_C) field of the Namespace Feature Descriptor; and

b) the `Assign` method specifies a NamespaceID parameter value that is equal to the NamespaceID column value in a Namespace Global Range Locking object,

then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

4.1.1.1.3.2.3 Preventing the assignment of an NNGRLO with an overlapping LBA range

Start of Informative Comment

The following rule prevents assignment of a Namespace Non-Global Range Locking object with an LBA range that overlaps the LBA range in another Namespace Non-Global Range Locking object for the same namespace/LUN. A zero-length LBA range (i.e., the RangeLength parameter is zero) does not overlap any LBA range, regardless of the value of the RangeStart parameter.

End of Informative Comment

If the `Assign` method specifies:

a) a NamespaceID parameter equal to the NamespaceID column value in any Namespace Non-Global Range Locking object; and

b) an LBA range in which one or more logical blocks are assigned to that Namespace Non-Global Range Locking object,

then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

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4.1.1.1.3.2.4 Preventing the assignment of an NNGRLO if no encryption key is available

Start of Informative Comment

The following rule prevents assignment of a Namespace Non-Global Range Locking object if there are no resources to store a new media encryption key.

End of Informative Comment

If:

a) the Storage Device reports a value of one in the Range Capable (Range_C) field of the Configurable Namespace Locking Feature Descriptor (see 5.2.1);

b) the NamespaceID parameter specifies a value that is equal to the NamespaceID column value in a Namespace Global Range Locking object;

c) the RangeStart parameter and the RangeLength parameter, if present, specify an LBA range in which no logical blocks are assigned to any Namespace Non-Global Range Locking object having a NamespaceID column value equal to the NamespaceID parameter; and

d) the Unused Key Count is equal to zero,

then the `Assign` method SHALL fail with a status of FAIL.

4.1.1.1.3.2.5 Preventing the assignment of more NNGRLOs than allowed

Start of Informative Comment

The following rule prevents assignment of more Namespace Non-Global Range Locking objects to a namespace/LUN than are indicated by the Maximum Ranges Per Namespace field of the Namespace Feature Descriptor.

End of Informative Comment

If:

a) the Maximum Ranges Per Namespace field of the Configurable Namespace Locking Feature Descriptor (see 5.2.1) specifies a value less than `0xFFFF_FFFF`;

b) the Unused Key Count is greater than zero; and

c) the value of the Maximum Ranges Per Namespace field is equal to the number of Namespace Non-Global Range Locking objects that have a NamespaceID column value equal to the NamespaceID parameter,

then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

4.1.1.1.3.3 General requirements

Start of Informative Comment

The following rule prevents the assignment of a Namespace Global Range Locking object or a Namespace Non-Global Range Locking object for a non-existent namespace/LUN. This rule implicitly prevents setting the NamespaceID column value to `0x0000_0000` or to `0xFFFF_FFFF` for Storage Devices that support the NVMe interface and `0xFFFF_FFFF` for Storage Devices that support the SCSI interface (see 4.2.2.1.1).

End of Informative Comment

If the Storage Device supports the NVMe interface and the NamespaceID parameter specifies a value that is not an allocated namespace identifier in the NVM subsystem [8], then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

If the Storage Device supports the SCSI interface and the NamespaceID parameter specifies a value of `0xFFFF_FFFF` or a LUN that does not exist, then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

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4.1.1.1.3.4 Additions to the namespace interactions specified in SIIS

The namespace interactions specified in SIIS ([3]) cover the Global LO / Multiple NS mode and the Multiple LO / Single NS mode shown in Figure 1.

Start of Informative Comment

The following rule describes the condition in which the Locking SP is in the Global LO / Multiple NS mode and is thus allowed to transition to the Multiple LO / Multiple NS mode. This condition is not the only condition in which the `Assign` method may be invoked.

End of Informative Comment

If:

a) the Storage Device supports the NVMe interface;

b) the TPer contains one or more namespaces; and

c) none of the Non-Global Range Locking objects is a configured locking object (as defined in section 2.7),

and if the `Assign` method succeeds with a status of SUCCESS, then the Locking SP SHALL transition to the Multiple LO / Multiple NS mode.

The cases in which the `Assign` method succeeds or fails are specified in 4.1.1.1.3 (`Assign` method operation) and in 5.6.2.1 (`Assign` method interactions with the Single User Mode Feature Set).

Start of Informative Comment

The following rule prevents successful operation of the `Assign` method if the Locking SP is in the Multiple LO / Single NS mode.

End of Informative Comment

If the Storage Device supports the NVMe interface, then, if the `Locking` table contains any Locking object for which:

a) the NamespaceID column value is zero or `0xFFFF_FFFF`; and

b) the locking object is a configured locking object (as defined in section 2.7),

then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

If the Storage Device supports the SCSI interface, then, if the `Locking` table contains any Locking object for which:

a) the NamespaceID column value is `0xFFFF_FFFF`; and

b) the locking object is a configured locking object (as defined in section 2.7),

then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

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4.1.1.1.3.5 Insufficient Unassigned Locking Objects

Start of Informative Comment

The following rule prevents successful operation of the `Assign` method when there are no suitable locking objects available for assignment.

End of Informative Comment

For the `Assign` method to succeed (terminate with a status code SUCCESS), the `Locking` table MUST contain at least one Non-Global Range Locking object that meets both the following NamespaceID column condition and the SUM mode matching condition, as defined below:

a) NamespaceID column condition:

   i. The Storage Device supports the NVMe interface and the locking object's NamespaceID column value is equal to zero or `0xFFFF_FFFF`, or:

   ii. The Storage Device supports the SCSI interface and the locking object's NamespaceID column value is equal to `0xFFFF_FFFF`.

b) SUM mode matching condition: The SUM mode of the locking object matches the AssignToSUMRange parameter, as defined below:

   A locking object is considered to match the SUM mode if:

   i. The locking object is included in the SingleUserModeRanges column and the AssignedToSUMRange parameter is set to TRUE; or

   ii. The locking object is not included in the SingleUserModeRanges column and the AssignedToSUMRange parameter is not specified or is set to FALSE.

If the `Locking` table does not contain any Non-Global Range Locking object that meets these conditions, then the `Assign` method SHALL fail with a status of INSUFFICIENT_ROWS (see [2]).

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4.1.1.2 Deassign (M)

The `Deassign` method is a Locking Template-specific method.

Start of Informative Comment

The `Deassign` method removes a Locking object's association with a namespace/LUN or namespace/LUN LBA range and resets multiple column values in a single operation.

The `Deassign` method is invoked upon a Locking SP which is in the Multiple LO / Multiple NS mode (see Figure 1).

End of Informative Comment

The following pseudo-code is the signature of the `Deassign` method (see [2] for more information).

```
LockingTableUID.Deassign[
    UID                          : uidref,
    KeepNamespaceGlobalRangeKey = boolean ]
=>
[ ]
```

Method UID: 00 00 00 06 00 00 08 05

4.1.1.2.1 Parameter Descriptions

4.1.1.2.1.1 UID

The UID parameter specifies the Locking object that is to be deassigned.

4.1.1.2.1.2 KeepNamespaceGlobalRangeKey

The KeepNamespaceGlobalRangeKey parameter specifies whether the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value is eradicated when a Namespace Global Range Locking object is deassigned.

A TRUE value for the KeepNamespaceGlobalRangeKey parameter is allowed only when the `Deassign` method is called on a Namespace Global Range Locking object.

4.1.1.2.2 Deassign Method Operation

Start of Informative Comment

The `Deassign` method is used to remove the association between an LBA range and a Namespace Non-Global Range Locking object or between a namespace/LUN and a Namespace Global Range Locking object. Logical blocks associated with the deassigned Locking object are then associated with another Locking object:

a) Deassigning a Namespace Non-Global Range Locking object:
   a. associates the logical blocks in the LBA range indicated by the Locking object with the Namespace Global Range Locking object associated with the namespace/LUN; and
   b. cryptographically erases the LBA range indicated by the Locking object.

b) Deassigning a Namespace Global Range Locking object:
   a. associates the logical blocks in the namespace/LUN with the Global Range Locking object; and
   b. cryptographically erases the LBA range indicated by the Locking object, if specified by the KeepNamespaceGlobalRangeKey parameter.

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The Deassign method is subject to constraints that prevent invalid configurations of the Locking SP. The following uses of Deassign are not allowed:

a) Deassigning a Namespace Global Range Locking object when there exist one or more Namespace Non-Global Range Locking objects associated with that namespace/LUN; and

b) Deassigning a Namespace Non-Global Range Locking object and retaining the media encryption key.

The constraint in a) above requires that the Deassign method be invoked successfully on all of the Namespace Non-Global Range Locking objects associated with the namespace/LUN before the Deassign method is invoked on the Namespace Global Range Locking object associated with the namespace/LUN.

It is required that the Namespace Global Range Locking object be deassigned from a namespace/LUN before the Namespace Management command is invoked to delete the namespace/LUN. For the details of using the Namespace Management command for this purpose, see 3.3.2 and [3].

End of Informative Comment

If the Deassign method succeeds, then:

1. the method SHALL set all column values in the selected Locking object to original factory values; and
2. if the Locking object is a Namespace Global Range Locking object, the method SHALL process the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value as specified by the KeepNamespaceGlobalRangeKey parameter.

4.1.1.2.2.1 Deassignment of an NNGRLO

Start of Informative Comment

The following rule specifies successful operation of the Deassign method on a Namespace Non-Global Range Locking object.

End of Informative Comment

If:

a) the Storage Device supports the NVMe interface;

b) the Locking object indicated by the UID parameter:

   a. is Read Unlocked and Write Unlocked;

   b. has the NamespaceGlobalRange column value set to FALSE; and

   c. has the NamespaceID column value set to a value other than zero or 0xFFFF_FFFF;

   and

c) the KeepNamespaceGlobalRangeKey parameter is FALSE,

then the Deassign method SHALL:

1. eradicate the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value;
2. increment the Unused Key Count by one; and
3. return a status code of SUCCESS.

If:

a) the Storage Device supports the SCSI interface;

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b) the Locking object indicated by the UID parameter:
   a. is Read Unlocked and Write Unlocked;
   b. has the NamespaceGlobalRange column value set to FALSE; and
   c. has the NamespaceID column value set to a value other than `0xFFFF_FFFF`;

   and

c) the KeepNamespaceGlobalRangeKey parameter is FALSE,

then the `Deassign` method SHALL:

1. eradicate the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value;
2. increment the Unused Key Count by one; and
3. return a status code of SUCCESS.

4.1.1.2.2.2 Preventing the deassignment of an NNGRLO if KeepNamespaceGlobalRangeKey parameter is TRUE

Start of Informative Comment

The following rule prevents successful operation of the `Deassign` method on a Namespace Non-Global Range Locking object with the KeepNamespaceGlobalRangeKey parameter set to TRUE.

End of Informative Comment

If:

a) the Locking object indicated by the UID parameter has the NamespaceGlobalRange column value set to FALSE; and

b) the KeepNamespaceGlobalRangeKey parameter is TRUE,

then the `Deassign` method SHALL fail with a status of INVALID_PARAMETER.

4.1.1.2.2.3 Preventing the deassignment of an NNGRLO if the locking object is locked

Start of Informative Comment

The following rule prevents deassignment of a Namespace Non-Global Range Locking object when that Locking object is Read Locked or Write Locked.

End of Informative Comment

If the Locking object indicated by the UID parameter:

a) has the NamespaceGlobalRange column value set to FALSE; and

b) is Read Locked or Write Locked,

then the `Deassign` method SHALL fail with a status of FAIL.

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4.1.1.2.2.4 Deassignment of an NGRLO, if KeepNamespaceGlobalRangeKey is TRUE

Start of Informative Comment

The following rule specifies successful operation of the `Deassign` method on a Namespace Global Range Locking object when the KeepNamespaceGlobalRangeKey parameter is set to TRUE.

End of Informative Comment

If:

1. the Global Range Locking object is Read Unlocked and Write Unlocked;
2. the Locking object indicated by the UID parameter:
    a. has the NamespaceGlobalRange column value set to TRUE;
    b. is Read Unlocked and Write Unlocked; and
    c. has a NamespaceID column value that is not equal to the NamespaceID column value in any other Locking object in the Locking SP;

    and
3. the KeepNamespaceGlobalRangeKey parameter is set to TRUE,

then the `Deassign` method:

1. SHALL NOT perform a cryptographic erase of the LBAs in that namespace/LUN;
2. SHALL NOT change the Unused Key Count;
3. SHALL transfer control of the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value of the Namespace Global Range Locking object to the K_AES_* object indicated by the ActiveKey column value of Global Range Locking object;
4. SHALL change the value of the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value of the Namespace Global Range Locking object; and
5. SHALL return a status code of SUCCESS.

4.1.1.2.2.5 Deassignment of an NGRLO, if KeepNamespaceGlobalRangeKey is FALSE

Start of Informative Comment

The following rule specifies successful operation of the `Deassign` method on a Namespace Global Range Locking object when the KeepNamespaceGlobalRangeKey parameter is set to FALSE.

End of Informative Comment

If:

a) the Global Range Locking object is Read Unlocked and Write Unlocked;
b) the Locking object indicated by the UID parameter:
    a. has the NamespaceGlobalRange column value set to TRUE;
    b. is Read Unlocked and Write Unlocked; and
    c. has a NamespaceID column value that is not equal to the NamespaceID column value in any other Locking object in the Locking SP;

    and

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c) the KeepNamespaceGlobalRangeKey parameter is set to FALSE,

then the `Deassign` method:

1. SHALL eradicate the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value;
2. SHALL fill in the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value of the Namespace Global Range Locking object with new key material;
3. SHALL transfer control of the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value of the Namespace Global Range Locking object to the K_AES_* object indicated by the ActiveKey column value of Global Range Locking object;
4. SHALL fill in the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value of the Namespace Global Range Locking object with new key material;
5. SHALL NOT change the Unused Key Count; and
6. SHALL return a status code of SUCCESS.

4.1.1.2.2.6 Preventing the deassignment of an NGRLO if its Namespace is associated with an NNGRLO

Start of Informative Comment

The following rule prevents deassignment of a Namespace Global Range Locking object when a Namespace Non-Global Range Locking object is associated with that namespace/LUN.

End of Informative Comment

If the Locking object indicated by the UID parameter:

a) has the NamespaceGlobalRange column value set to TRUE; and

b) has a NamespaceID column value equal to the NamespaceID column value in another assigned Locking object in the Locking SP,

then the `Deassign` method SHALL fail with a status of INVALID_PARAMETER.

4.1.1.2.2.7 Preventing the deassignment of an NGRLO if either the NGRLO or the GRLO are locked

Start of Informative Comment

The following rule prevents deassignment of a Namespace Global Range Locking object when that Locking object or the Global Range Locking object is Read Locked or Write Locked.

End of Informative Comment

If:

a) the Locking object indicated by the UID parameter:
   a. has the NamespaceGlobalRange column value set to TRUE; and
   b. is Read Locked or Write Locked;

   or

b) the Global Range Locking object is Read Locked or Write Locked,

then the `Deassign` method SHALL fail with a status of FAIL.

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4.1.1.2.2.8 Error if the Deassign method is applied to the GRLO or to a non-exsistent or unassigned NGRLO

Start of Informative Comment

The following rule specifies the error to report when the `Deassign` method is invoked on a non-existent or unassigned Namespace Global Range Locking object. The rule also specifies the error to report when the `Deassign` method is invoked on the Global Range Locking object.

End of Informative Comment

If the Storage Device supports the NVMe interface and the Locking object indicated by the UID parameter:

a) does not exist; or

b) has a NamespaceID column value of zero or `0xFFFF_FFFF`,

then the `Deassign` method SHALL fail with a status of INVALID_PARAMETER.

If the Storage Device supports the SCSI interface and the Locking object indicated by the UID parameter:

a) does not exist; or

b) has a NamespaceID column value of `0xFFFF_FFFF`,

then the `Deassign` method SHALL fail with a status of INVALID_PARAMETER.

4.1.2 Modified Methods

This feature set modifies the following methods:

a) `Set`;

b) `Revert`; and

c) `RevertSP`.

4.1.2.1 Set

Start of Informative Comment

The `Set` method is subject to constraints that prevent invalid configurations of the Locking SP. The following uses of `Set` are not allowed, based on access control settings:

a) Changing which namespace/LUN is associated with a Locking object (i.e., changing the NamespaceID column value);

b) Changing a Namespace Global Range Locking object into a Namespace Non-Global Range Locking object (i.e., changing the NamespaceGlobalRange column value);

c) Changing a Namespace Non-Global Range Locking object into a Namespace Global Range Locking object (i.e., changing the NamespaceGlobalRange column value); and

d) Assigning more than one Namespace Global Range Locking object for the same namespace/LUN.

Because the constraints described in this section apply to Namespace Global Range Locking objects and Namespace Non-Global Range Locking objects, they are meaningful only when the Locking SP is in the Multiple LO / Multiple NS mode (see 3.1).

The constraints described in this section are enforced by the access control list for the Locking SP, which does not permit the `Set` method to modify either the NamespaceID column value or NamespaceGlobalRange column value.

End of Informative Comment

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4.1.2.1.1 Preventing the Set method from causing LBA ranges to overlap

Start of Informative Comment

The following rule prevents the modification of a Namespace Non-Global Range Locking object to cause its LBA range to overlap the LBA range of another Namespace Non-Global Range Locking object.

End of Informative Comment

If the `Set` method is invoked on a Namespace Non-Global Range Locking object and specifies an LBA range in which one or more logical blocks are associated with a different Non-Global Range Locking object specifying the same NamespaceID column value, then the `Set` method SHALL fail with a status of INVALID_PARAMETER.

4.1.2.1.2 Preventing the Set method from modifying RangeStart and RangeLength of an NGRLO

Start of Informative Comment

The following rule prevents the modification of the RangeStart and RangeLength columns on Namespace Global Range Locking objects.

End of Informative Comment

If the `Set` method is invoked on a Namespace Global Range and specifies values for the RangeStart or RangeLength columns, then the `Set` method SHALL fail with a status of INVALID_PARAMETER.

4.1.2.1.3 Interactions with the namespace management model specified in SIIS

Start of Informative Comment

The namespace management model specified in SIIS applies when the Locking SP is in either the Global LO / Multiple NS mode or the Multiple LO / Single NS mode (see 3.1), i.e., the `Locking` table does not contain a Namespace Global Range Locking object or a Namespace Non-Global Range Locking object. The Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set functionality is disallowed when the Locking SP is in the Global LO / Multiple NS mode.

End of Informative Comment

If the Storage Device supports the NVMe interface and any Locking object has a non-zero NamespaceID column value and the `Set` method is invoked on any Locking object with a NamespaceID column value of zero, other than the Global Range Locking object, then the `Set` method SHALL fail with a status of INVALID_PARAMETER.

4.1.2.2 Revert

Upon a successful invocation of the `Revert` method that results in reverting the Locking SP, the method SHALL increment the Unused Key Count by the number of Namespace Non-Global Range Locking objects when the method was invoked.

The Unused Key Count SHALL NOT be otherwise affected by the `Revert` method invocation, i.e. the Unused Key Count SHOULD NOT be returned to its OFS values.

4.1.2.3 RevertSP

Start of Informative Comment

If a namespace is associated with a Namespace Global Range Locking object when the `RevertSP` method is invoked, then the media encryption key of that namespace is eradicated, regardless of whether the KeepGlobalRangeKey parameter is set to TRUE or FALSE.

If the invoker of the `RevertSP` method wishes to keep the keys associated with Namespace Global Range Locking objects, the `Deassign` method should be invoked on those Namespace Global Range Locking objects prior to invoking the `RevertSP` method.

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If the Deassign method is successfully invoked on a Namespace Global Range Locking object before the RevertSP method is invoked, then the media encryption key of that namespace/LUN would be associated with the Global Range Locking object at time of RevertSP method invocation.

End of Informative Comment

Upon successful invocation of the RevertSP method, the method SHALL increment the Unused Key Count by the number of Namespace Non-Global Range Locking objects when the method was invoked.

If the RevertSP method is invoked with the KeepGlobalRangeKey parameter set to TRUE, then the TPer SHALL:

a) continue to use the media encryption key for each namespace/LUN that was associated with the Global Range Locking object; and

b) eradicate the media encryption key associated with the K_AES_* object indicated by the ActiveKey column value of each Non-Global Locking object.

The Unused Key Count SHALL NOT be otherwise affected by the RevertSP method invocation, i.e. the Unused Key Count SHOULD NOT be returned to its OFS values.

4.2 Tables

This section defines new tables and modifications to existing tables required for this feature set.

4.2.1 New Tables

There are no new tables defined by this feature set.

4.2.2 Modified Tables

This feature set adds columns to the Locking SP's Locking table.

This feature set adds rows to the Locking SP's MethodID, ACE, and AccessControl tables' preconfiguration, and modifies rows of these tables.

The preconfiguration modifications are described in 5.4.1.

4.2.2.1 Locking SP

This feature set modifies the Locking table by adding the following columns (see Table 5), in addition to those defined in [2]:

Table 5 - Locking SP – Locking Table Columns

Column Number	Column Name	IsUnique	Column Type
0x14	NamespaceID	See Note 1	bytes_4
0x15	NamespaceGlobalRange	See Note 1	boolean
Note 1: The content of the cell is defined by the Core Specification [2] and the applicable Opal Family SSC.			

The behavior of the Global Range Locking object is modified (see 5.4.1.1.1).

4.2.2.1.1 NamespaceID (M)

Start of Informative Comment

The NamespaceID column name was defined prior to SCSI LUNs being included in this feature set. The name was retained to maintain backward compatibility.

End of Informative Comment

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The NamespaceID column value indicates which namespace/LUN is associated with this Locking object.

If the Storage Device supports the NVMe interface, the NamespaceID column value of the Global Range Locking object SHALL be set to 0x0000_0000 or `0xFFFF_FFFF`.

If the Storage Device supports the SCSI interface, the NamespaceID column value of the Global Range Locking object SHALL be set to `0xFFFF_FFFF`.

If the Storage Device supports the SCSI interface, the Storage Device SHALL map LUN ID values (eight bytes) to NamespaceID column values (4 bytes) by taking the value of the first four bytes (i.e., 0-3) of the LUN ID.

The bytes-ordering of NamespaceID column is Big-Endian.

4.2.2.1.2 NamespaceGlobalRange (M)

Start of Informative Comment

The NamespaceGlobalRange column name was defined prior to SCSI LUNs being included in this feature set. The name was kept to maintain backward compatibility.

End of Informative Comment

The NamespaceGlobalRange column value indicates whether the Locking object is associated with:

a) a designated LBA range in the namespace/LUN; or

b) all logical blocks in the namespace/LUN that are not associated with any other Locking object.

If the NamespaceGlobalRange column value is FALSE, then the Locking object is associated with the LBA range indicated by the RangeStart and RangeLength column values, in the namespace/LUN indicated by the NamespaceID column value.

If the NamespaceGlobalRange column value is TRUE, then the Locking object is associated with all logical blocks in the namespace/LUN that are not associated with any other Locking object.

The NamespaceGlobalRange column value of the Global Range Locking object SHALL be ignored.

4.3 Types

This section defines new types and modifications to existing types required for this feature set.

4.3.1 New Types

There are no new types defined by this feature set.

4.3.2 Modified Types

There are no types modified by this feature set.

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5 Feature Set Requirements

This section defines the Mandatory (M) and Optional (O) requirements for the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set.

5.1 Requirements Overview

The Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set consists of namespace/LUN specific capabilities that MAY be implemented in a TPer. A Host discovers the namespace/LUN specific capabilities and properties of a TPer by examining the Configurable Namespace Locking Feature Descriptor returned in Level 0 Discovery.

5.2 Level 0 Discovery

A Storage Device implementing the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set SHALL:

a) return the Configurable Namespace Locking Feature Descriptor as defined in 5.2.1; and

b) support the Level 0 Discovery response requirements defined in the various Opal Family SSCs.

5.2.1 Configurable Namespace Locking Feature Descriptor (Feature Code = 0x0403) (M)

The Configurable Namespace Locking Feature Descriptor SHALL be returned when the Storage Device supports the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set. The contents of the feature descriptor are defined in Table 6.

Table 6 - Level 0 Discovery – Configurable Namespace Locking Feature Descriptor

Byte \ Bit	7	6	5	4	3	2	1	0
0	(MSB)			Feature Code				
1								(LSB)
2	Feature Descriptor Version Number				Feature Set Minor Version Number			
3	Length							
4	Range_C	Range_P	SUM_C	Reserved				
5 – 7	Reserved							
8	(MSB)			Maximum Key Count				
…								
11								(LSB)
12	(MSB)			Unused Key Count				
…								
15								(LSB)
16	(MSB)			Maximum Ranges Per Namespace				
…								
19								(LSB)

5.2.1.1 Feature Code

The Feature Code field value SHALL be set to `0x0403`.

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5.2.1.2 Feature Descriptor Version Number

The Feature Descriptor Version Number field SHALL be set to `0x2`.

5.2.1.3 Feature Set Minor Version Number

The Feature Set Minor Version Number represents the minor version of the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set supported by the Storage Device.

The Feature Set Minor Version Number field SHALL be set to a value as specified in Table 7.

Table 7 - Feature Set Minor Versions

Feature Set Minor Version Number	Standard Referenced
0x0	Configurable Namespace Locking v1.00
0x1	Configurable Locking for NVMe Namespaces and SCSI LUNs v1.01
0x2	Configurable Locking for NVMe Namespaces and SCSI LUNs v1.02
0x3	Configurable Locking for NVMe Namespaces and SCSI LUNs v1.30
All others	Reserved

5.2.1.4 Length

The Length field indicates the number of bytes in the descriptor following byte 3. The value of the Length field SHALL be set to `0x10`.

5.2.1.5 Range_C

The Range Capable (Range_C) field is set to one to indicate that the Storage Device supports Namespace Non-Global Range Locking objects. The Range_C field is set to zero to indicate that the Storage Device does not support Namespace Non-Global Range Locking objects.

If the Range_C field is set to one, then the Storage Device SHALL support Namespace/LUN Level 0 Discovery as defined in 5.2.2.

If the Range_C field is set to zero, then the Storage Device SHOULD support Namespace/LUN Level 0 Discovery as defined in 5.2.2.

5.2.1.6 Range_P

The Range Present (Range_P) field is set to one to indicate that the `Locking` table contains one or more Namespace Non-Global Range Locking objects. The Range_P field is set to zero to indicate that the `Locking` table does not contain any Namespace Non-Global Range Locking objects.

5.2.1.7 SUM_C

The SUM Capable (SUM_C) field indicates whether the SD supports assigning namespaces and LBA ranges within namespaces to ranges that are activated in Single User Mode.

If the SUM_C field is set to one, then the SD SHALL support the AssignToSUMRange parameter of the `Assign` method (see section 4.1.1.1.1.4) to facilitate assigning namespaces and LBA ranges within namespaces to ranges that are activated in Single User Mode (see section 5.6.2.1.2).

If the SUM_C field is set to one, then the SD SHALL support the Single User Mode Feature Set defined in [9].

If the SUM_C field is cleared to zero, then the SD SHALL NOT support the AssignToSUMRange parameter of the `Assign` method and does not allow assigning namespaces and LBA ranges within namespaces to ranges that are activated in Single User Mode.

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5.2.1.8 Maximum Key Count

The Maximum Key Count field indicates the maximum number of media encryption keys the Storage Device supports concurrently.

The value of the Maximum Key Count field SHALL be set during the Storage Device manufacturing process.

For NVMe, this value SHALL be greater than or equal to the maximum number of namespaces supported by the NVM subsystem (see [8]).

For SCSI, this value SHALL be greater than or equal to the maximum number of LUNs supported by the Storage Device. (see [10]).

The value of the Maximum Key Count field MAY be less than the Number of Namespaces multiplied by the value of the Maximum Ranges Per Namespace field.

5.2.1.9 Unused Key Count

The Unused Key Count field indicates how many media encryption keys are unused and are thus available for use.

If the Locking SP is in the Multiple LO / Single NS mode, the value of the Maximum Key Count field and the value of the Unused Key Count field are related according to the following equation:

```
Unused Key Count   =  Maximum Key Count
                        - (  number of existing namespaces/LUNs in the TPer (i.e., one)
                           +  number of configured Non-Global Range Locking objects
                          )
```

If the Locking SP is in the Global LO / Multiple NS mode or in the Multiple LO / Multiple NS mode, the value of the Maximum Key Count field and the value of the Unused Key Count field are related according to the following equation:

```
Unused Key Count   =  Maximum Key Count
                        - ( number of existing namespaces/LUNs in the TPer
                           + number of configured Namespace Non-Global Range Locking objects
                          )
```

Start of Informative Comment

It is the host's responsibility to consider the number of available keys (Unused Key Count) when calling commands and methods that consume keys (such as the Namespace Management command and the `Assign` method). Otherwise, these commands and methods may fail.

End of Informative Comment

5.2.1.10 Maximum Ranges Per Namespace

The Maximum Ranges Per Namespace field indicates the maximum number of Namespace Non-Global Range Locking objects that can be configured for each namespace/LUN. If the Maximum Ranges Per Namespace field is set to a value less than `0xFFFF_FFFF`, then that value is the maximum number of Namespace Non-Global Range Locking objects that are able to be assigned to any namespace/LUN.

If the Maximum Ranges Per Namespace field is set to `0xFFFF_FFFF`, then a limit set by this field SHALL NOT apply.

If the Range_C field is set to one, then the Maximum Ranges Per Namespace field SHALL be set to a nonzero value.

If the Range_C field is set to zero, then the Maximum Ranges Per Namespace field SHALL be set to zero.

The value indicated by this field does not dynamically change when a TPer's LockingSP transitions out of the "Multiple LO / Multiple NS" mode (see section 3.1 for details on the LockingSP modes transitions).

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5.2.2 Namespace/LUN Level 0 Discovery

In this section, the terms “NVMe NamespaceID (NSID)” and “SCSI LUN” refer to the entity identifiers at the underlying transport layer.

5.2.2.1 Overview

The Namespace/LUN Level 0 Discovery command provides a host with basic information about TPer capabilities both current and potential, for a specific Namespace or LUN.

5.2.2.2 IF-SEND Command

There is no IF-SEND command defined for Namespace/LUN Level 0 Discovery, so if IF-SEND is invoked with a Protocol ID of `0x01` and a ComID of `0x0002`, then the TPer SHALL:

1. transfer all of the data from the host;
2. discard the data; and
3. return ’good’ status to the host.

5.2.2.3 IF-RECV Command

Namespace/LUN Level 0 Discovery is invoked by sending an IF-RECV command with:

Protocol ID = `0x01`
ComID = `0x0002`
Transfer Length = maximum length of the Namespace/LUN Level 0 Discovery response data that the host elects to receive.

The Namespace/LUN Level 0 Discovery IF-RECV command MAY be processed at any time, without regard to sessions or prior authentication.

If the Transfer Length parameter is less than the size of the Namespace/LUN Level 0 Discovery response data that is available, then the TPer SHALL return the requested amount of data, even if it is truncated.

If the Transfer Length parameter is greater than the size of the Namespace/LUN Level 0 Discovery response data, then the device SHALL pad according to the rules specified in the transport.

If NVMe NSID or SCSI LUN specify a value that corresponds to an existing namespace/LUN (i.e., an NVMe allocated Namespace Identifier (NSID) [8] or a SCSI LUN identifier that is not an incorrect logical unit number [10]), then the TPer SHALL return Namespace/LUN Level 0 Discovery Response Data containing feature descriptors corresponding to that namespace/LUN.

If NVMe NSID or SCSI LUN specify a value of `0xFFFF_FFFF`, then the TPer SHALL return the Namespace/LUN Level 0 Discovery header and no feature descriptors.

If NVMe NSID or SCSI LUN is any other value, then the TPer SHALL fail the command with a status of Other Invalid Command Parameter.

The Namespace/LUN Level 0 Discovery response data (see Table 8) consists of a header field and zero or more variable length feature descriptors. A TPer SHALL NOT include feature descriptors for namespace/LUN features that it does not implement. The data is not packetized.

Table 8 – Namespace/LUN Level 0 Discovery Response Data Format

Bit / Byte	7	6	5	4	3	2	1	0
0 – 47	Namespace/LUN Level 0 Discovery Header (see Table 9)							
48 – n	Feature Descriptor(s) (see 5.2.2.4)							

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The Namespace/LUN Level 0 Discovery Header format is defined in Table 9.

Table 9 – Namespace/LUN Level 0 Discovery Header Format

Bit / Byte	7	6	5	4	3	2	1	0
0	(MSB)							
…				Length of Parameter Data				
3								(LSB)
4	(MSB)							
…				Data Structure Revision				
7								(LSB)
8								
…				Reserved				
47								

5.2.2.3.1 Length of Parameter Data

The Length of the Parameter Data field indicates the total number of bytes that are valid in the Namespace/LUN Level 0 Discovery header and all of the feature descriptors returned, not including this field. If no feature descriptors are returned, then this field SHALL be set to 0x0000_002C.

5.2.2.3.2 Data Structure Revision

The Data Structure Revision field describes the format of the Namespace/LUN Level 0 Discovery header returned. The value SHALL be 0x0000_0001.

5.2.2.4 Namespace/LUN Level 0 Discovery Feature Descriptors

The Namespace/LUN Feature Descriptors SHALL be returned in the Namespace/LUN Level 0 Discovery response data in order of increasing namespace/LUN feature code values. Namespace/LUN features that are not implemented SHALL NOT be returned.

Table 10 contains the feature code for Namespace/LUN Level 0 Discovery.

Table 10 – Namespace/LUN Level 0 Discovery Feature Codes

Feature Code	Feature Name	Description
0x0405	Namespace Geometry Reporting Feature	See 5.2.2.5

5.2.2.5 Namespace Geometry Reporting Feature (Feature Code = 0x0405)

The Namespace Geometry Reporting Feature (see Table 11) indicates the logical block and physical block geometry supported within the namespace/LUN indicated by the NVMe NSID or SCSI LUN parameter of the IF-RECV command. The Namespace Geometry Reporting Feature Descriptor feature MAY be returned in the Namespace/LUN Level 0 Discovery response. See [2] for additional information.

Table 11 – Namespace/LUN Level 0 Discovery – Namespace Geometry Reporting Feature Descriptor

Bit / Byte	7	6	5	4	3	2	1	0
0	(MSB)			Feature Code				
1								(LSB)
2	Feature Descriptor Version Number				Reserved			
3	Length							
4	Reserved							Align
5	(MSB)	Reserved						

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Bit Byte	7	6	5	4	3	2	1	0
…								
11								(LSB)
12	(MSB)							
…				LogicalBlockSize				
15								(LSB)
16	(MSB)							
…				AlignmentGranularity				
23								(LSB)
24	(MSB)							
…				LowestAlignedLBA				
31								(LSB)

5.2.2.5.1 Feature Code
The Feature Code field SHALL be set to `0x0405`.

5.2.2.5.2 Feature Descriptor Version Number
The Feature Descriptor Version Number field SHALL be set to `0x1`.

5.2.2.5.3 Length
The Length field indicates the number of bytes in the descriptor following byte 3. The value SHALL be set to `0x1C`.

5.2.2.5.4 Align
The Align field indicates whether the TPer requires ranges in the specified namespace/LUN to be aligned. If Align is TRUE then the TPer requires ranges in the specified Namespace/LUN to be aligned. If Align is FALSE, then the TPer does not require ranges to be aligned.

5.2.2.5.5 LogicalBlockSize
The LogicalBlockSize field indicates the number of bytes in a logical block for the specified Namespace/LUN.

5.2.2.5.6 AlignmentGranularity
The AlignmentGranularity field indicates the number of logical blocks in a group, for alignment purposes within the specified Namespace/LUN (For details about access granularity see the applicable Opal Family SSC).

5.2.2.5.7 LowestAlignedLBA
The LowestAlignedLBA field indicates the lowest logical block address that is located at the beginning of an alignment granularity group for the specified Namespace/LUN (For details about access granularity see the applicable Opal Family SSC).

5.3 Admin SP

This feature set modifies the behavior of the `Revert` method (see 4.1.2.2).

5.4 Locking SP

A Storage Device implementing this feature set SHALL support the additions to the Locking SP specified in this section, in addition to the Locking SP requirements defined in the applicable Opal Family SSC.

This feature set modifies the behavior of the `RevertSP` method (see 4.1.2.3).

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5.4.1 Tables

5.4.1.1 Locking Table (M)

5.4.1.1.1 Global Range Locking Object (M)

The Global Range Locking object SHALL apply to all namespaces/LUNs that are not associated with any other Locking object.

If the Storage Device supports the NVMe interface, the NamespaceID column value of the Global Range Locking object SHALL be set to `0x0000_0000` or `0xFFFF_FFFF`.

If the Storage Device supports the SCSI interface, the NamespaceID column value of the Global Range Locking object SHALL be set to `0xFFFF_FFFF`.

The NamespaceGlobalRange column value of the Global Range Locking object SHALL be set to TRUE.

5.4.1.1.2 Locking Table Preconfiguration

Section 4.2.2.1 specifies two additional columns in the `Locking` Table: NamespaceID and NamespaceGlobalRange.

In addition to the requirements in the applicable Opal Family SSC, these added columns SHALL be preconfigured as specified in Table 12.

The conventions used by Table 12 are defined in section 2.3.2. Superscript numbers within table cells refer to notes at the bottom of the table.

Table 12 - Locking SP Locking Table Preconfiguration of Added Columns

UID	Name	NamespaceID	NamespaceGlobalRange
00 00 08 02 00 00 00 01	“Locking_GlobalRange” [1]	`0x0000_0000`	**T**
00 00 08 02 00 03 00 01	“Locking_Range1” [1]	`0x0000_0000` [2]	**F** [2]
00 00 08 02 00 03 NN NN	“Locking_RangeNNNN” [1]	`0x0000_0000` [2]	**F** [2]

Notes:
1. The text in the table cell is informative. See the Core Specification [2] and the applicable Opal Family SSC.
2. The cell is indirectly writeable using the `Assign` method and the `Deassign` method.

5.4.1.2 MethodID Table (M)

5.4.1.2.1 MethodID Table Preconfiguration

In addition to the requirements in the applicable Opal Family SSC, the `MethodID` table preconfiguration SHALL be modified as specified in Table 13.

The conventions used by Table 13 are defined in section 2.3.2. Superscript numbers within table cells refer to notes at the bottom of the table.

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Table 13 - Locking SP MethodID Table Preconfiguration – Added Rows

UID	Name	CommonName	TemplateID
00 00 00 06 00 00 08 04	"Assign" [1]	See note 2	See note 2
00 00 00 06 00 00 08 05	"Deassign" [1]	See note 2	See note 2

Notes:
1. The text in the table cell is informative. See the Core Specification [2] and the applicable Opal Family SSC.
2. The content of the cell is defined by the Core Specification [2] and the applicable Opal Family SSC.

5.4.1.3 AccessControl Table (M)

5.4.1.3.1 AccessControl Table Preconfiguration

In addition to the requirements in the applicable Opal Family SSC, the `AccessControl` table SHALL be preconfigured as specified in Table 14.

The conventions used by Table 14 are defined in section 2.3.2. Superscript numbers within table cells refer to notes at the bottom of the table.

Table 14 - Locking SP AccessControl Table Preconfiguration

Table Association – informative only	UID	InvokingID	InvokingID Name – informative only	MethodID	CommonName	ACL	Log	AddACEACL	RemoveACEACL	GetACLACL	DeleteMethodACL	AddACELog	RemoveACELog	GetACLLog	DeleteMethodLog	LogTo
Locking																
	(ND)[3]	00 00 08 02 00 00 00 00	LockingTableUID [1]	Assign	See note 2	ACE_Assign	(NR)[4]	(NR)[4]	(NR)[4]	ACE_Anybody	(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]
	(ND)[3]	00 00 08 02 00 00 00 00	LockingTableUID [1]	Deassign	See note 2	ACE_Deassign	(NR)[4]	(NR)[4]	(NR)[4]	ACE_Anybody	(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]

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		Table Association – informative only
(ND)[3]	(ND)[3]	**UID**
00 00 08 02 00 03 00 01	00 00 08 02 00 00 00 01	**InvokingID**
Locking_Range1 [1]	Locking_GlobalRange [1]	InvokingID Name – informative only
Get	Get	**MethodID**
See note 2	See note 2	**CommonName**
ACE_Locking_Range1_Get_RangeStartToActiveKey, ACE_Anybody_Get_CommonName, ACE_Locking_Namespace_IDtoGlbRng	ACE_Locking_GlobalRange_Get_RangeStartToActiveKey, ACE_Anybody_Get_CommonName, ACE_Locking_Namespace_IDtoGlbRng	**ACL**
(NR)[4]	(NR)[4]	**Log**
(NR)[4]	(NR)[4]	**AddACEACL**
(NR)[4]	(NR)[4]	**RemoveACEACL**
ACE_Anybody	ACE_Anybody	**GetACLACL**
(NR)[4]	(NR)[4]	**DeleteMethodACL**
(NR)[4]	(NR)[4]	**AddACELog**
(NR)[4]	(NR)[4]	**RemoveACELog**
(NR)[4]	(NR)[4]	**GetACLLog**
(NR)[4]	(NR)[4]	**DeleteMethodLog**
(NR)[4]	(NR)[4]	**LogTo**

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	ACE		Table Association – informative only
(ND)[3]		(ND)[3]	**UID**
00 00 00 08 00 03 80 02		00 00 08 02 00 03 00 00 (+NN NN)	**InvokingID**
ACE_Locking_Namespace_IdtoGlbRng [1]		Locking_RangeNNNN [1]	InvokingID Name – informative only
Set		Get	**MethodID**
See note 2		See note 2	**CommonName**
ACE_ACE_Set_BooleanExpression		ACE_Locking_RangeNNNN_Get_RangeStartToActiveKey, ACE_Anybody_Get_CommonName, ACE_Locking_Namespace_IDtoGlbRng	**ACL**
(NR)[4]		(NR)[4]	**Log**
(NR)[4]		(NR)[4]	**AddACEACL**
(NR)[4]		(NR)[4]	**RemoveACEACL**
ACE_Anybody		ACE_Anybody	**GetACLACL**
(NR)[4]		(NR)[4]	**DeleteMethodACL**
(NR)[4]		(NR)[4]	**AddACELog**
(NR)[4]		(NR)[4]	**RemoveACELog**
(NR)[4]		(NR)[4]	**GetACLLog**
(NR)[4]		(NR)[4]	**DeleteMethodLog**
(NR)[4]		(NR)[4]	**LogTo**

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				Table Association – informative only
(ND)[3]	(ND)[3]	(ND)[3]	(ND)[3]	**UID**
00 00 00 08 00 03 F9 02	00 00 00 08 00 03 F9 01	00 00 00 08 00 03 F9 01	00 00 00 08 00 03 80 02	**InvokingID**
ACE_Deassign [1]	ACE_Assign [1]	ACE_Assign [1]	ACE_Locking_Namespace_IdtoGlbRng [1]	InvokingID Name – informative only
Get	Set	Get	Get	**MethodID**
See note 2	See note 2	See note 2	See note 2	**CommonName**
ACE_ACE_Get_All	ACE_ACE_Set_BooleanExpression	ACE_ACE_Get_All	ACE_ACE_Get_All	**ACL**
(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	**Log**
(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	**AddACEACL**
(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	**RemoveACEACL**
ACE_Anybody	ACE_Anybody	ACE_Anybody	ACE_Anybody	**GetACLACL**
(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	**DeleteMethodACL**
(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	**AddACELog**
(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	**RemoveACELog**
(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	**GetACLLog**
(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	**DeleteMethodLog**
(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	**LogTo**

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Table Association – informative only	**UID**	**InvokingID**	InvokingID Name – informative only	**MethodID**	**CommonName**	**ACL**	**Log**	**AddACEACL**	**RemoveACEACL**	**GetACLACL**	**DeleteMethodACL**	**AddACELog**	**RemoveACELog**	**GetACLLog**	**DeleteMethodLog**	**LogTo**
	(ND)[3]	00 00 00 08 00 03 F9 02	ACE_Deassign [1]	Set	See note 2	ACE_ACE_Set_BooleanExpression	(NR)[4]	(NR)[4]	(NR)[4]	ACE_Anybody	(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]	(NR)[4]

Notes:
1. The text in the cell is informative.
2. The content of the cell is defined by the Core Specification [2] and the applicable Opal Family SSC.
3. The content of the cell is not defined.
4. The content of the cell is not required. See the Core Specification [2] and the applicable Opal Family SSC.

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5.4.1.4 ACE Table (M)

5.4.1.4.1 ACE Table Preconfiguration

In addition to the requirements in the applicable Opal Family SSC, the ACE table SHALL be preconfigured as specified in Table 15.

The conventions used by Table 15 are defined in section 2.3.2. Superscript numbers within table cells refer to notes at the bottom of the table.

Table 15 - Locking SP ACE Table Preconfiguration

Table Association – informative only	UID	Name	CommonName	BooleanExpr	Columns
Locking					
	00 00 00 08 00 03 80 02	"ACE_Locking_Namespace_IdtoGlbRng" [1]	See note 2	Admins	NamespaceID, NamespaceGlobalRange
ACE					
	00 00 00 08 00 03 F9 01	“ACE_Assign” [1]	See note 2	Admins	All
	00 00 00 08 00 03 F9 02	“ACE_Deassign” [1]	See note 2	Admins	All

Notes:
1. The text in the table cell is informative. See the Core Specification [2] and the applicable Opal Family SSC.
2. The content of the cell is defined by the Core Specification [2] and the applicable Opal Family SSC.

5.5 Additional SPs

This feature set requires no additional SPs.

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5.6 Single User Mode Feature Set Interactions

5.6.1 Overview

The Single User Mode Feature Set (see [9]) MAY be supported on a TPer that supports the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set.

This section describes the interactions when the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set and the Single User Mode Feature Set (see [9]) are both supported and present (enabled or disabled).

User authorities that are Single User Mode Locking object owners SHALL NOT be permitted to be added to ACE_Assign and ACE_Deassign.

Start of Informative Comment

The Single User Mode Feature Set adds columns that are referenced by this feature set:

- `LockingInfo` table new column: SingleUserModeRanges: A list of UIDs that identify the Locking objects that are in Single User Mode; or the `Locking` table UID if all ranges are in Single User Mode.
- `LockingInfo` table new column: RangeStartLengthPolicy: An enumeration value that identifies the management policy for the RangeStart, RangeLength, and CommonName columns of Locking objects that are in Single User Mode:
  - A value of 0 defines that the `Locking` table columns RangeStart, RangeLength and CommonName are owned by the locking object's associated User authority.
  - A value of 1 defines that the `Locking` table columns RangeStart, RangeLength and CommonName are owned by the Admins authority.

End of Informative Comment

5.6.2 Modified Methods

This section defines modifications to methods that are required to support the Single User Mode Feature Set (see [9]) in addition to the Configurable Locking for NVMe Namespaces and SCSI LUNs Feature Set.

5.6.2.1 Assign

`Assign` method interactions with Single User Mode vary based on the column value of the RangeStartLengthPolicy column in the `LockingInfo` table.

5.6.2.1.1 Assign interactions when RangeStartLengthPolicy is 0

5.6.2.1.1.1 Preventing the assignment of an NGRLO if the GRLO is under exclusive control of a user authority

Start of Informative Comment

The following rule prevents assignment of a Namespace Global Range Locking object if the Global Locking object is under exclusive control of a user authority.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the NamespaceID parameter specifies a value that is not equal to the NamespaceID column value of any Locking object;

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c) the RangeStartLengthPolicy column value of the LockingInfo table is 0; and

d) the Global Range Locking object is included in the SingleUserModeRanges column of the LockingInfo table,

then the Assign method SHALL fail with a status of NOT_AUTHORIZED.

5.6.2.1.1.2 Preventing the assignment of an NNGRLO if the NGRLO is under exclusive control of a user authority

Start of Informative Comment

The following rule prevents assignment of a Namespace Non-Global Range Locking object if the Namespace Global Locking object is under exclusive control of a user authority.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the RangeStartLengthPolicy column value of the LockingInfo table is 0; and

c) the NamespaceID parameter specifies a value that is equal to the NamespaceID column value in a Namespace Global Range Locking object that is included in the SingleUserModeRanges column of the LockingInfo table,

then the Assign method SHALL fail with a status of NOT_AUTHORIZED.

5.6.2.1.2 Assign interactions when RangeStartLengthPolicy is 1

5.6.2.1.2.1 Assigning an NGRLO if the GRLO is activated in Single User Mode

Start of Informative Comment

The following rule defines the assignment of a Namespace Global Range Locking object when the Global Range Locking object is included in the SingleUserModeRanges column of the LockingInfo table and the AssignToSUMRange is not specified or is set to FALSE.

Compared to the normal Assign operation, assigning a namespace associated with the Global Range when the Global Range is activated in Single User Mode should result in erasure of data in the namespace as part of the Assign operation.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the NamespaceID parameter specifies a value that is not equal to the NamespaceID column value of any Locking object;

c) the RangeStartLengthPolicy column value of the LockingInfo table is 1;

d) the UID of the Global Range Locking object is included in the SingleUserModeRanges column of the LockingInfo table; and

e) the AssignToSUMRange parameter is not specified or is set to FALSE,

then the Assign method SHALL succeed as defined in section 4.1.1.1.3.1 and the SD SHALL perform the following actions:

1. eradicate the media encryption key associated with the namespace; and

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2. transfer control of the media encryption key associated with the namespace to the K_AES_* object indicated by the ActiveKey column value of the selected Locking object.

5.6.2.1.2.2 Assignment of an NGRLO that is activated in Single User Mode using AssignToSUMRange

Start of Informative Comment

The following rule defines the assignment of a Namespace Global Range Locking object that is activated in Single User Mode using AssignToSUMRange.

Compared to the normal Assign operation, assigning a namespace to a Namespace Global Range Locking object that is activated in Single User Mode should result in erasure of data in the namespace as part of the Assign operation.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the NamespaceID parameter specifies a value that is not equal to the NamespaceID column value of any Locking object;

c) the RangeStartLengthPolicy column value of the `LockingInfo` table is 1; and

d) the AssignToSUMRange parameter is set to TRUE,

then the `Assign` method SHALL succeed as defined in section 4.1.1.1.3.1 and the SD SHALL perform the following actions:

1. Select a Non-Global Range Locking object that:
   a. has a NamespaceID column value of zero; and
   b. has a UID that is included in the SingleUserModeRanges column of the `LockingInfo` table.
2. eradicate the media encryption key associated with the namespace; and
3. transfer control of the media encryption key associated with the namespace to the K_AES_* object indicated by the ActiveKey column value of the selected Locking object.

5.6.2.1.2.3 Assignment of an NNGRLO if the NGRLO is activated in Single User Mode

Start of Informative Comment

The following rule defines the assignment of a Namespace Non-Global Range Locking object when the Namespace Global Range Locking object associated with the namespace is included in the SingleUserModeRanges column of the `LockingInfo` table and the AssignToSUMRange is not specified or is set to FALSE.

Assigning a namespace associated with a Namespace Global Range Locking object activated in Single User Mode should result in erasure of data in the LBA range specified in the `Assign` method.

The Core Specification [2] states:
“If the RangeStart/RangeLength column values are returned to their previous values, it is not guaranteed that the original data is preserved on the logical blocks whose control transitioned from one LBA range to another”.

This rule relies on this behavior to erase data when assigning Namespace Non-Global Range Locking object.

End of Informative Comment

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If:

a) the method would otherwise succeed;

b) the NamespaceID parameter specifies a value that is equal to the NamespaceID column value of a Namespace Global Locking object;

c) the RangeStartLengthPolicy column value of the `LockingInfo` table is 1;

d) The Namespace Global Range Locking object associated with the namespace is included in the SingleUserModeRanges column of the `LockingInfo` table; and

f) the AssignToSUMRange parameter is not specified or is set to FALSE,

then the `Assign` method SHALL succeed as defined in section 4.1.1.1.3.2.

5.6.2.1.2.4 Assignment of an NNGRLO that is activated in Single User Mode using AssignToSUMRange

Start of Informative Comment

The following rule defines the assignment of a Namespace Non-Global Range Locking object that is activated in Single User Mode using AssignToSUMRange.

Assigning a namespace to a Namespace Non-Global Range Locking object activated in Single User Mode should result in erasure of data in the LBA range specified in the `Assign` method.

The Core Specification [2] states:

"If the RangeStart/RangeLength column values are returned to their previous values, it is not guaranteed that the original data is preserved on the logical blocks whose control transitioned from one LBA range to another".

This rule relies on this behavior to erase data when assigning Namespace Non-Global Range Locking object that is activated in Single User Mode.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the NamespaceID parameter specifies a value that is equal to the NamespaceID column value of a Namespace Global Locking object;

c) the RangeStartLengthPolicy column value of the `LockingInfo` table is 1; and

d) the AssignToSUMRange paramater is set to TRUE,

then the `Assign` method SHALL succeed as defined in section 4.1.1.1.3.2 and the SD SHALL perform the following actions:

1. Select a Non-Global Range Locking object that:
   a. has a NamespaceID column value of zero; and
   b. has a UID that is included in the SingleUserModeRanges column of the `LockingInfo` table.

5.6.2.1.2.5 Error condition with conflicting use of the AssignToSUMRange parameter

Start of Informative Comment

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The following rule defines the error condition when the `Assign` method is invoked with the AssignToSUMRange parameter set to TRUE and SUM is not supported.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the SD does not support the Single User Mode Feature Set (see [9]); and

c) the AssignToSUMRange parameter is set to TRUE,

then the `Assign` method SHALL fail with a status of INVALID_PARAMETER.

5.6.2.2 Deassign

`Deassign` method interactions with Single User Mode vary based on the column value of the RangeStartLengthPolicy column in the `LockingInfo` table.

5.6.2.2.1 Deassign interactions when RangeStartLengthPolicy is 0

Start of Informative Comment

The following rule prevents deassignment of a Locking object if that Locking object is under exclusive control of a user authority.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the RangeStartLengthPolicy column value of the `LockingInfo` table is 0; and

c) the value of the UID parameter is included in the SingleUserModeRanges column of the `LockingInfo` table,

then the `Deassign` method SHALL fail with a status of NOT_AUTHORIZED.

5.6.2.2.2 Deassign interactions when RangeStartLengthPolicy is 1

5.6.2.2.2.1 Deassigning an NGRLO that is activated in Single User Mode if KeepNamespaceGlobalRangeKey is not specified or set to FALSE

Start of Informative Comment

The following rule defines deassignment of a Namespace Global Range Locking object that is activated in Single User Mode and KeepNamespaceGlobalRangeKey is not specified or set to FALSE.

Deassigning a Namespace Global Range Locking object with KeepNamespaceGlobalRangeKey not specified or set to FALSE results in erasure of the data associated with the Namespace Global Locking object.

The Core Specification [2] states:
“If the RangeStart/RangeLength column values are returned to their previous values, it is not guaranteed that the original data is preserved on the logical blocks whose control transitioned from one LBA range to another”.

This rule relies on this behavior to erase data when Deassigning a Namespace Global Range Locking object that is activated in Single User Mode.

End of Informative Comment

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If:

a) the method would otherwise succeed;

b) the RangeStartLengthPolicy column value of the `LockingInfo` table is 1;

c) the Locking object indicated by the UID parameter is a Namespace Global Range Locking object;

d) the value of the UID parameter of the `Deassign` method is included in the SingleUserModeRanges column of the `LockingInfo` table; and

e) the KeepNamespaceGlobalRangeKey parameter is:

   a. Not specified; or

   b. Set to FALSE,

then the `Deassign` method SHALL succeed as defined in section 4.1.1.2.2.

5.6.2.2.2.2 Deassigning an NNGRLO that is activated in Single User Mode

Start of Informative Comment

The following rule defines deassignment of a Namespace Non-Global Range Locking object that is activated in Single User Mode.

Deassigning a Namespace Non-Global Range Locking object is similar to de-personalizing a non-global locking range which results in erasure of the data associated with the non-global locking range.

The Core Specification [2] states:
“If the RangeStart/RangeLength column values are returned to their previous values, it is not guaranteed that the original data is preserved on the logical blocks whose control transitioned from one LBA range to another”.

This rule relies on this behavior to erase data when Deassigning a Namespace Non-Global Range Locking object that is activated in Single User Mode.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the RangeStartLengthPolicy column value of the `LockingInfo` table is 1;

c) the Locking object indicated by the UID parameter is a Namespace Non-Global Range Locking object; and

d) the value of the UID parameter of the `Deassign` method is included in the SingleUserModeRanges column of the `LockingInfo` table,

then the `Deassign` method SHALL succeed as defined in section 4.1.1.2.2.

5.6.2.2.2.3 Preventing the deassignment of an NGRLO that is activated in SUM if KeepNamespaceGlobalRangeKey is TRUE

Start of Informative Comment

The following rule prevents deassignment of a Namespace Global Range Locking object that is activated in Single User Mode when KeepNamespaceGlobalRangeKey is TRUE.

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The objective is to prevent other authorities from being able to transition ownership of the namespace associated with the Single User Mode back to the global range without erasing data.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the RangeStartLengthPolicy column value of the `LockingInfo` table is 1;

c) the Locking object indicated by the UID parameter is a Namespace Global Range Locking object;

d) the value of the UID parameter of the `Deassign` method is included in the SingleUserModeRanges column of the `LockingInfo` table; and

e) the KeepNamespaceGlobalRangeKey parameter is TRUE,

then the `Deassign` method SHALL fail with a status of NOT_AUTHORIZED.

5.6.2.2.2.4 Preventing the deassignment of an NGRLO if the GRLO is activated in SUM and KeepNamespaceGlobalRangeKey is TRUE

Start of Informative Comment

The following rule prevents deassignment of a Namespace Global Range Locking object when the Global Range Locking object is activated in Single User Mode and KeepNamespaceGlobalRangeKey is TRUE.

The objective is to prevent other authorities from being able to transition ownership of the namespace to the global range activated in Single User Mode without erasing data.

End of Informative Comment

If:

a) the method would otherwise succeed;

b) the RangeStartLengthPolicy column value of the `LockingInfo` table is 1;

c) the Locking object indicated by the UID parameter is a Namespace Global Range Locking object;

d) the UID of the Global Range Locking object is included in the SingleUserModeRanges column of the `LockingInfo` table; and

e) the KeepNamespaceGlobalRangeKey parameter is TRUE,

then the `Deassign` method SHALL fail with a status of NOT_AUTHORIZED.

5.6.2.3 Reactivate Method

Upon successful invocation of the `Reactivate` method:

a) The NamespaceID and NamespaceGlobalRange column values SHALL remain at their current values.

b) User authorities that are owners of Single User Mode Locking objects SHALL be removed from the BooleanExpr column values of the ACE_Assign and the ACE_Deassign ACEs.

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5.6.2.4 Set Method

Start of Informative Comment

The following rule prevents the addition of Single User Mode Locking object owners to the ACE_Assign or ACE_Deassign ACEs

End of Informative Comment

If:

a) the `Set` method is invoked on the ACE_Assign or the ACE_Deassign ACEs;

b) the BooleanExpr column is included in the `Set` method parameters; and

c) the value of the BooleanExpr column parameter includes one or more User authorities that are Single User Mode Locking object owners,

then the TPer SHALL fail the `Set` method invocation with status INVALID_PARAMETER.

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